MAT121 MATERIAL QUESTIONS (1) AND SOLUTIONS. (3 unit)

Solved and Compiled by Sheriff a.k.a Schrödinger

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INTRODUCTION TO CALCULUS

(1) LIMIT OF A FUNCTION
- Substitution method
- Factorization method
- L-hopital rule

(2) DIFFERENTIATION
- product rule
- chain rule
- Quotient rule or Division rule
- implicit function
- Trigonometry Function
- inverse Of Trigonometry
- logarithm Functions
- Exponential Function
- Hyperbolic Function
- Higher Derivative
- Application of Differentiation
- Maximum and Minimum point

③ INTEGRATION

→ Constant of integration
→ Integration of polynomial
→ Standard integral
→ trigonometry integral
→ exponential integral
→ integration of Forms i.e

$$= \frac{f'(x)}{f(x)}\,dx$$

$$= f(x) \cdot f'(x)\,dx$$

→ integration by parts

Terms under limit

① $\frac{1}{\infty} = 0$

② $\frac{1}{0} = \infty$

③ $\frac{1}{0^2} = \infty$

④ $\frac{0}{0}$ = undefined

⑤ $\frac{\infty}{\infty}$ = undefined

⑥ $\frac{5}{\infty} = 0$

⑦ anything divided by ∞ = 0

Symbol ∞ = infinity

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Limit of a function by Substitution methods

Examples

(1) $\lim_{x \to 2} (x^2 + 2x + 3)$

Solution

$(2)^2 + 2(2) + 3$

$4 + 4 + 3 = 11$ //

(2) $\lim_{x \to 0} (x^4 + x + 4)$

solution

$(0)^4 + (0) + 4$

$0 + 0 + 4 = 4$ //

(3) $\lim_{x \to 1} (x^5 - 2x^3 + 1)$

solution

$(1)^5 - 2(1)^3 + 1$

$1 - 2 + 1$

$-1 + 1 = 0$ //

(4) $\lim_{x \to 3} (x^2 - x - 2)$

Solution

$(3)^2 - 3 - 2$

$9 - 3 - 2$

$= 6 - 2$

$= 4$

(5) $\lim_{h \to -2} (h^2 - 2h - 4)$

Solution

$(-2)^2 - 2(-2) - 4$

$4 + 4 - 4$

$8 - 4$

$= 4$

(6) $\lim_{x \to a} (x^3 - 4x^2 + x)$

$= (a)^3 - 4(a)^2 + a$

$a^3 - 4a^2 + a$

11. Current economic and social issues.

(10) $\lim_{x \to 0} (x^2 + 4x)$

$= (0)^2 + 4(0)$

$0 + 0 = 0$ //

(11) $\lim_{x \to \infty} (x^2 - 3x + 1)$

$= \infty^2 - 3\infty + 1$

$= \infty$

(because anything plus $\infty = \infty$
anything minus $\infty = \infty$)

(12) $\lim_{x \to \infty} \left(\frac{1}{x}\right)$

$= \frac{1}{\infty} = 0$

(13) $\lim_{x \to \infty} \left(\frac{1}{x^2}\right)$

$= \frac{1}{\infty^2} = 0$

(14) $\lim_{x \to \infty} \left(\frac{1}{x^3}\right)$

$\frac{1}{\infty^3} = 0$

Similarly

(anything multiply by infinity (∞) $= \infty$)

$\infty \times 2 = \infty$

$\infty + 5 = \infty$

$\infty + 100 = \infty$

$\infty + 1000 = \infty$

$\infty - 10 = \infty$

$\infty \times 500 = \infty$

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(7) $\lim_{m \to 0} (m^6 - 2m - 6)$

$= (0)^6 - 2(0) - 6$

$0 - 0 - 6$

$= -6$ //

(8) $\lim_{x \to 2} \left(\frac{4x+3}{x+5}\right)$

$= \frac{4(2)+3}{2+5} = \frac{8+3}{7}$

$= \frac{11}{7}$ //

(9) $\lim_{x \to 1} \left(\frac{x^3+3x+1}{x^2+4+1}\right)$

$= \frac{(1)^3 + 3(1) + 1}{(1)^2 + 4 + 1}$

$= \frac{1+3+1}{1+4+1} = \frac{5}{6} = \frac{5}{6}$ //

8 //

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(15) $\lim_{x \to 0} \left(\frac{1}{x}\right)$

$= \frac{1}{0} = \infty$

(16) $\lim_{x \to 0} \left(\frac{1}{x^5}\right)$

$= \frac{1}{0^5} = \infty$

(17) $\lim_{x \to 0} \left(\frac{1}{x^{500}}\right)$

$\frac{1}{0^{500}} = \infty$

(18) $\lim_{x \to \infty} \left(x^2 + 3 + x^4\right)$

$= (\infty)^2 + 3 + (\infty)^4$

$\infty + 3 + \infty$

$= \infty$ //

(19) $\lim_{x \to \infty} \left(x^3 + 3x^4\right)$

$= (\infty)^3 + 3(\infty)^4$

$\infty + \infty = \infty$ //

(20) $\lim_{x \to 5} \left(x^3 + 4x^2 + 2x + 1\right)$

$= (5)^3 + 4(5)^2 + 2(5) + 1$

$5^3 + 4(5)^2 + 2(5) + 1$

Limit of a function by factorization methods

Examples

(1) $\lim_{x \to 2} \left(\frac{x^2 - 4}{x-2} \right)$

$= x^2 - 4$

$= (x-2)(x+2)$

$\lim_{x \to 2} \left(\frac{(x-2)(x+2)}{(x-2)} \right)$

$\lim_{x \to 2} \left(\frac{x+2}{1} \right) = \frac{2+2}{1}$

$= \frac{4}{1} = 4$ //

(2) $\lim_{x \to 3} \left(\frac{x^2 - 9}{x-3} \right)$

$= x^2 - 9 = (x-3)(x+3)$

$\lim_{x \to 3} \left(\frac{(x-3)(x+3)}{x-3} \right)$

$\lim_{x \to 3} \left(\frac{x+3}{1} \right) = \frac{3+3}{1} = \frac{6}{1}$

$= 6$ //

(3) $\lim_{x \to 4} \left(\frac{x^2 - 16}{x-4} \right)$

$= x^2 - 16 = (x-4)(x+4)$

$\lim_{x \to 4} \left(\frac{(x-4)(x+4)}{(x-4)} \right)$

$\left(\frac{x+4}{1} \right) = \frac{4+4}{1} = 8$ //

$= 8$ //

(4) $\lim_{x \to 5} \left(\frac{x^2 - 25}{x-5} \right)$

$\left(\frac{(x-5)(x+5)}{(x-5)} \right) = \frac{(x+5)}{1}$

$5+5 \Rightarrow 10/1 =$

10 //

(5) $\lim_{x \to 6} \left(\frac{x^2 - 36}{x - 6} \right)$

$(x^2 - 36) = (x - 6)(x + 6)$

$\lim_{x \to 6} \left(\frac{(x-6)(x+6)}{(x-6)} \right) = \left(\frac{x+6}{1} \right)$

$\lim_{x \to 6} \left(\frac{x+6}{1} \right) = \frac{6+6}{1} = \frac{12}{1} = 12$ //

(6) $\lim_{h \to 7} \left(\frac{h^2 - 49}{h - 7} \right)$

$(h^2 - 49) = (h - 7)(h + 7)$

$\lim_{h \to 7} \left(\frac{(h-7)(h+7)}{(h-7)} \right) = \left(\frac{h+7}{1} \right) = \frac{7+7}{1} = \frac{14}{1}$

$= 14$ //

(7) $\lim_{h \to 8} \left(\frac{h^2 - 64}{h - 8} \right)$

$(h^2 - 64) = (h - 8)(h + 8)$

$\lim_{h \to 8} \left(\frac{(h-8)(h+8)}{(h-8)} \right)$

$\lim_{h \to 8} \left(\frac{h+8}{1} \right) = \frac{8+8}{1} = \frac{16}{1} = 16$ //

(8) $\lim_{x \to 9} \left(\frac{x^2 - 81}{x - 9} \right)$

$x^2 - 81 = (x - 9)(x + 9)$

$\lim_{x \to 9} \left(\frac{(x-9)(x+9)}{(x-9)} \right) = \left(\frac{x+9}{1} \right)$

$\Rightarrow \frac{9+9}{1} = \frac{18}{1} = 18$ //

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(9) $\lim_{b \to 10} \left\{ \frac{b^2 - 100}{b - 10} \right\}$

$b^2 - 100 = (b-10)(b+10)$

$\lim_{b \to 10} \left(\frac{(b-10)(b+10)}{(b-10)} = \frac{b+10}{1} \right.$

$\frac{10+10}{1} = \frac{20}{1} = 20$ //

(10) $\lim_{h \to 0} \left(\frac{(x+h)^2 - x^2}{h} \right)$

we expand this $(x+h)^2$ using binomial expansion

$(x+h)^2 = x^2h + x^1h^1 + x^0h^2$

$x^2 + xh + h^2$

from pascal triangle

1

1 1

1 2 1

$x^2 + 2xh + h^2$

Substitute $x^2 + 2xh + h^2$ into the main question given

$$\lim_{h\to 0}\left(\frac{x^2+2xh+h^2-x^2}{h}\right) \Rightarrow \left(\frac{2xh+h^2}{h}\right)$$

factorize h

$$\lim_{h\to 0} h\left(\frac{2x+h}{h\times 1}\right)$$

$$= \frac{2x+h}{h}$$

Put the value of $h \to 0$

$$\frac{2x+0}{1} = \frac{2x}{1} = 2x$$

$$\underline{\underline{2x}}$$

by Shortcut approach

$$\lim_{h\to 0}\left(\frac{(x+h)^2-x^2}{h}\right)$$

Just differentiate x^2

i.e $x^2 = 2x$

ignoring the minus (-) sign

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(11) $\lim_{h\to 0}\left(\frac{(x+h)^3-x^3}{h}\right)$

we expand $(x+h)^3$ using binomial expansion

$(x+h)^3 = x^3h^0 + x^2h^1 + x^1h^2 + x^0h^3$

$= x^3 + x^2h + xh^2 + h^3$

1
1 1
1 2 1
1 3 3 1

$x^3 + 3x^2h + 3xh^2 + h^3$

Substitute $x^3+3x^2h+3xh^2+h^3$ into the question given

$\lim_{h\to 0}\left(\frac{x^3+3x^2h+3xh^2+h^3 - x^3}{h}\right) = \left(\frac{3x^2h+3xh^2+h^3}{h}\right)$

factorize h

$\lim_{h\to 0} \not{h}\left(\frac{3x^2+3xh+h^2}{\not{h}\times 1}\right) = \frac{3x^2+3xh+h^2}{1}$

Put the value of $h \to 0$

$\frac{3x^2+3x(0)+(0)^2}{1} = \frac{3x^2+0+0}{1} = \frac{3x^2}{1}$

$= 3x^2$

Short cut approach

$$\lim_{h\to 0}\left(\frac{(x+h)^3 - x^3}{h}\right)$$

Just differentiate x^3

i.e $x^3 = 3x^2$

ignoring the minus (-) sign

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(12) $$\lim_{h\to 0}\left(\frac{(x+h)^4 - x^4}{h}\right)$$

we expand $(x+h)^4$ using binomial expansion

$(x+h)^4 = x^4h^0 + x^3h^1 + x^2h^2 + x^1h^3 + x^0h^4$

$x^4 + x^3h + x^2h^2 + xh^3 + h^4$

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
```

$(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$

Substitute $x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$ into the given function

$$\lim_{h\to 0}\left(\frac{x^4+4x^3h+6x^2h^2+4xh^3+h^4-x^4}{h}\right)$$

$$\lim_{h\to 0}\left(\frac{4x^3h+6x^2h^2+4xh^3+h^4}{h}\right)$$

factorize h

$$h\left(\frac{4x^3+6x^2h+4xh^2+h^3}{h\times 1}\right)$$

$$\frac{4x^3+6x^2h+4xh^2+h^3}{1}$$

put $h\to 0$

$$\frac{4x^3+6x^2(0)+4x(0)^2+(0)^3}{1}=\frac{4x^3+0+0+0}{1}$$

$$\frac{4x^3}{1}=4x^3$$

$$=4x^3$$

by shortcut approach

differentiate x^4 and ignoring the minus (−) sign

i.e $\lim_{h\to 0}\left(\frac{(x+h)^4-x^4}{h}\right)$

$$x^4=4x^3$$

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(13) $\lim_{h \to 0} \left(\frac{(x+h)^5 - x^5}{h} \right)$

$(x+h)^5 = x^5h^0 + x^4h^1 + x^3h^2 + x^2h^3 + x^1h^4 + x^0h^5$

$x^5 + x^4h + x^3h^2 + x^2h^3 + xh^4 + h^5$

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

$(x+h)^5 = x^5 + 5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^4$

$\lim_{h \to 0} \left(\frac{x^5 + 5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5 - x^5}{h} \right)$

$\lim_{h \to 0} \left(\frac{5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5}{h} \right)$

$h \left(\frac{5x^4 + 10x^3h + 10x^2h^2 + 5xh^3 + h^4}{h \times 1} \right)$

$\frac{5x^4 + 10x^3h + 10x^2h^2 + 5xh^3 + h^4}{1}$

at $h \to 0$

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(17)

$$\lim_{h\to 0}\left(\frac{5x^4+10x^3(0)+10x^2(0)^2+5x(0)^3+(0)^5}{1}\right)$$

$$\frac{5x^4+0+0+0+0}{1} = \frac{5x^4}{1}$$

$$= 5x^4$$

by shortcut approach

$$\lim_{h\to 0}\left(\frac{(x+h)^5-x^5}{h}\right)$$

differentiate x^5 and ignoring the minus (-) sign

$$x^5 = 5x^4$$

(14)

$$\lim_{h\to 0}\left(\frac{(x+h)^6-x^6}{h}\right)$$

by short cut approach

differentiate x^6

$$x^6 = 6x^5$$

(15) $\lim_{h \to 0} \left(\frac{(x+h)^7 - x^7}{h} \right)$

$=$

Just differentiate x^7

$x^7 = 7x^6$

(16) $\lim_{h \to 0} \left(\frac{(x+h)^8 - x^8}{h} \right)$

Just differentiate x^8

$x^8 = 8x^7$

(17) $\lim_{h \to 0} \left(\frac{(x+h)^9 - x^9}{h} \right)$

Just differentiate x^9

i.e $x^9 = 9x^8$

10. National income
11. Current economic and social issues

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(18) $\lim_{x \to 4} \left(\frac{x-4}{x^2-x-12} \right)$

factorize $\Rightarrow$

$\Rightarrow x^2 - x - 12$

$= x^2 - 4x + 3x - 12$

$= x^2 - 4x + 3x - 12$

$x(x-4) + 3(x-4)$

$(x+3)(x-4)$

$\lim_{x \to 4} \left(\frac{x-4}{(x+3)(x-4)} \right) = \left(\frac{1}{x+3} \right)$

$\lim_{x \to 4} \left(\frac{1}{4+3} \right) = \frac{1}{7}$ //

(19) $\lim_{x \to 11} \left(\frac{x^2 - 121}{x-11} \right)$

$(x^2 - 121) = (x-11)(x+11)$

$\left(\frac{(x-11)(x+11)}{x-11} \right) \Rightarrow \frac{x+11}{1} = \frac{11+11}{1} = \frac{22}{1}$

$= 22$

(20)

(20)

$$\lim_{x \to 12} \left(\frac{x^2 - 144}{x - 12} \right) \Rightarrow (x^2 - 144) = (x - 12)(x + 12)$$

$$\left(\frac{(x-12)(x+12)}{x-12} \right) = \left(\frac{x+12}{1} \right)$$

$$\lim_{x \to 12} = \frac{12+12}{1} = \frac{24}{1} = 24$$

limit of a function by L-hopital rule

In this case, we first substitute the value in which x tend to when it gives you zero or infinity (∞), We differentiate the numerator and denominator separately and put the value which x tend to

Examples

(1) $\lim_{x \to 4} \left(\frac{x - 4}{x^2 - 3x - 12} \right)$

$$\Rightarrow \left(\frac{4-4}{(4)^2 - 4 - 12} \right) = \frac{0}{16-16} = \frac{0}{0} = \text{undefined}$$

undefined

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$$\lim_{x \to 4} \left(\frac{x-4}{x^2-x-12} \right)$$

$$= \frac{1-0}{2x-1-0}$$

$$\frac{1}{2x-1}$$

put $x \to 4$

$$\frac{1}{2x4-1} = \frac{1}{8-1} = \frac{1}{7} //$$

(2) $$\lim_{x \to 2} \left(\frac{x^2-4}{x-2} \right)$$

$$\Rightarrow \frac{(2)^2-4}{2-2} = \frac{4-4}{2-2} = \frac{0}{0}$$

= undefined

$$\lim_{x \to 2} \left(\frac{x^2-4}{x-2} \right)$$

$$= \frac{2x-0}{1-0} \Rightarrow \frac{2x}{1}$$

Put $x \to 2$

$$\frac{2(2)}{1} = \frac{4}{1} = 4$$

(3) $$\lim_{x \to 3} \left(\frac{x^2-9}{x-3} \right)$$

$$\Rightarrow \frac{(3)^2-9}{3-3} = \frac{9-9}{33} = \frac{0}{0}$$

undefined

$$\lim_{x \to 3} \left(\frac{x^2-9}{x-3} \right)$$

$$\frac{2x-0}{1-0}$$

$$= \frac{2x}{1}$$

$$\frac{2(3)}{1} = 6 //$$

(4) $\lim_{x \to 4} \left(\frac{x^2-16}{x-4}\right)$

$\frac{(4)^2-16}{4-4} \Rightarrow \frac{16-16}{4-6}$

$\Rightarrow \frac{0}{0}$ = undefined

$\lim_{x \to 4} \left(\frac{x^2-16}{x-4}\right)$

$\frac{2x-0}{1-0} = \frac{2x}{1}$

$\frac{2(4)}{1} = 8//$

$= 8$

(5) $\lim_{x \to 5} \left(\frac{x^2-25}{x-5}\right)$

$\Rightarrow \frac{5^2-25}{5-5} = \frac{25-25}{5-5}$

$\frac{0}{0}$ = undefined

$\lim_{x \to 5} \left(\frac{x^2-25}{x-5}\right)$

$\frac{2x-0}{1-0}$

$\frac{2x}{1} = \frac{2(5)}{1} = 10/1$

$= 10//$

(6) $\lim_{x \to 6} \left(\frac{x^2-36}{x-6}\right)$

$\frac{6^2-36}{6-6} = \frac{36-36}{6-6}$

$\frac{0}{0}$ = undefined

$\left(\frac{x^2-36}{x-6}\right) = \frac{2x-0}{1}$

$\frac{2(6)}{1} = 12//$

(7) $\lim_{x \to 1} \left(\frac{x^3 - 1}{x - 1} \right)$

$\frac{(1)^3 - 1}{1 - 1} = \frac{1 - 1}{1 - 1} = \frac{0}{0}$

= undefined

$\lim_{x \to 1} \left(\frac{x^3 - 1}{x - 1} \right)$

$\frac{3x^2 - 0}{1 - 0} = \frac{3x^2}{1}$

put $x \to 1$

$\frac{3(1)^2}{1} = 3$

(8) $\lim_{x \to m} \left(\frac{x^2 + mx - 2m^2}{x^2 - m^2} \right)$

$= \frac{m^2 + m(m) - 2m^2}{m^2 - m^2}$

$\frac{m^2 + m^2 - 2m^2}{m^2 - m^2}$

$\frac{2m^2 - 2m^2}{m^2 - m^2} = \frac{0}{0}$ = undefined

$\lim_{x \to m} \left(\frac{x^2 + mx - 2m^2}{x^2 - m^2} \right)$

$\frac{2m + m - 0}{2x - 0}$

$\frac{2m + m}{2x} \Rightarrow$

put $x \to m$

$\frac{2m + m}{2m} = \frac{3x}{2x}$

$= \frac{3}{2}$

(9) $\lim_{\theta \to 0} \left(\frac{\sin \theta}{\theta} \right)$

$\Rightarrow \frac{\sin(0)}{0} = \frac{0}{0}$

undefined

$$\lim_{\theta \to 0} \left(\frac{\sin\theta}{\theta} \right)$$

$$\frac{\cos\theta}{1}$$

Put $\theta \to 0$

$$\frac{\cos(0)}{1} = \frac{1}{1} = 1 //$$

(10) $$\lim_{x \to 0} \left(\frac{1-\cos x}{x} \right)$$

$$\frac{1-\cos(0)}{0}$$

$$\frac{1-1}{0} = \frac{0}{0}$$ undefined

$$\lim_{x \to 0} \left(\frac{1-\cos x}{x} \right)$$

$$\frac{0+\sin x}{1}$$

$$\frac{0+\sin(0)}{1} = \frac{0}{1} = 0 //$$

Hence, the ans = 0 //

Hence, the answer = 0 //

(11) $$\lim_{x \to \frac{\pi}{2}} \left(\frac{\sin x + x}{x} \right)$$

$$= \frac{\sin\frac{180^\circ}{2} + \frac{180^\circ}{2}}{\frac{180^\circ}{2}}$$

because

$$\frac{\pi}{2} = \frac{180^\circ}{2}$$

$$\pi = 180^\circ$$

$$\frac{\sin 90^\circ + 90^\circ}{90}$$

$$\sin 90^\circ = 1$$

$$\frac{1+90^\circ}{90} = \frac{91}{90} //$$

(12) $\lim_{x \to 1} \left(\frac{x^4 - 1}{x - 1} \right)$

$= \frac{(1)^4 - 1}{1 - 1}$

$= \frac{1 - 1}{1 - 1} = \frac{0}{0}$ = undefined

$\lim_{x \to 1} \left(\frac{x^4 - 1}{x - 1} \right)$

$\frac{4x^3 - 0}{1 - 0}$

$\frac{4x^3}{1}$

Put $x \to 1$

$\frac{4(1)^3}{1} = 4$ //

(13) $\lim_{x \to 7} \left(\frac{x^2 - 49}{x - 7} \right)$

$\frac{(7)^2 - 49}{7 - 7} = \frac{49 - 49}{7 - 7}$

$\frac{0}{0}$ = undefined

$\lim_{x \to 7} \left(\frac{x^2 - 49}{x - 7} \right)$

$\frac{2x - 0}{1 - 0} = \frac{2x}{1}$

Put $x \to 7$

$\frac{2(7)}{1} = 14$ //

(14) $\lim_{x \to 8} \left(\frac{x^2 - 64}{x - 8} \right)$

$\frac{8^2 - 64}{8 - 8} = \frac{0}{0}$ = undefined

$\left(\frac{x^2 - 64}{x - 8} \right)$

$\frac{2x - 0}{1 - 0} = \frac{2x}{1}$

Put $x \to 8$

$\frac{2(8)}{1} = 16$ //

(15) $$\lim_{x \to 4} \frac{16(\sqrt{x}) - x^{5/2}}{4-x}$$

$$= \frac{16 \cdot x^{1/2} - x^{5/2}}{4-x}$$

differentiating numerator and denominator

$$= \frac{16 \times \frac{1}{2} x^{\frac{1}{2}-1} - \frac{5}{2} x^{\frac{5}{2}-1}}{0-1}$$

$$= \frac{\frac{16}{2} x^{-1/2} - \frac{5}{2} x^{3/2}}{-1}$$

$$= \frac{8x^{-1/2} - \frac{5}{2} x^{3/2}}{1}$$

but

$x^{-1} = \frac{1}{x}$

$x^{-1/2} = \frac{1}{x^{1/2}}$

$$\frac{\frac{8}{x^{1/2}} - \frac{5}{2}(x^{3/2})}{-1}$$

$$\frac{\frac{8}{\sqrt{x}} - \frac{5}{2}(\sqrt{x^3})}{-1}$$

put $x \to 4$

$$\frac{\frac{8}{\sqrt{4}} - \frac{5}{2}(\sqrt{4^3})}{-1}$$

$$\frac{\frac{8}{2} - \frac{5}{2}(\sqrt{64})}{-1}$$

$$\frac{\frac{8}{2} - \frac{5}{2}(8)}{-1}$$

because

$\sqrt{64} = 8$

$$\frac{4 - \frac{40}{2}}{-1}$$

$$\frac{4 - 20}{-1}$$

$$-(4-20)$$

$$= -4+20$$

$$= 16$$

limit To infinity

Recall that

$\frac{1}{0} = \infty$, $\frac{1}{\infty} = 0$

(Anything multiply, Add, Substract by Infinity(∞) is equal $= \infty$)

$2 + \infty = \infty$

$5 + \infty = \infty$

$50 \times \infty = \infty$

$50 - \infty = \infty$

$10 + \infty = \infty$

$1 - \infty = \infty$

$$\lim_{x \to 0} \left(\frac{1}{x^2}\right) = \frac{1}{0^2} = \frac{1}{0} = \infty$$

$$\lim_{x \to \infty} \left(\frac{1}{x^5}\right) = \frac{1}{\infty^5} = \frac{1}{\infty} = 0$$

Note: limit to infinity is done or solve by dividing the numerator and denominator by the variable with the highest power in the functions

Examples

① $\lim_{x \to \infty} \left(\frac{4x+3}{2x+1} \right)$

divide numerator and the denominator by x

$$\frac{\frac{4x}{x} + \frac{3}{x}}{\frac{2x}{x} + \frac{1}{x}}$$

$$= \frac{4 + 3/x}{2 + \frac{1}{x}}$$

put the value which $x \to$ to ∞

$$\frac{4 + 3/\infty}{2 + 1/\infty}$$

$$\frac{4 + 0}{2 + 0} = 4/2 //$$

$$= 2 //$$

② $\lim_{x \to \infty} \left(\frac{2x^2 + 3x + 1}{4x^2 + 2x + 2} \right)$

= divide numerator and denominator by highest power i.e x^2

$$= \frac{\frac{2x^2}{x^2} + \frac{3x}{x^2} + \frac{1}{x^2}}{\frac{4x^2}{x^2} + \frac{2x}{x^2} + \frac{2}{x^2}}$$

$$= \frac{2 + \frac{3}{x} + \frac{1}{x^2}}{4 + \frac{2}{x} + \frac{2}{x^2}}$$

put $x \to \infty$

$$\frac{2 + \frac{3}{\infty} + \frac{1}{\infty^2}}{4 + \frac{2}{\infty} + \frac{2}{\infty^2}}$$

$$\frac{2 + 0 + 0}{4 + 0 + 0}$$

$$\frac{2}{4} = \frac{1}{2}$$

$$= \frac{1}{2} //$$

Solved and Compiled by Sheriff a.k.a Schrödinger (29)

(3) $\lim_{x\to\infty}\left(\frac{x^3+2x^2+5}{5x^3+3x^2+1}\right)$

= divide numerator and denumerator by highest power i.e x^3

$$= \frac{\frac{x^3}{x^3}+\frac{2x^2}{x^3}+\frac{5}{x^3}}{\frac{5x^3}{x^3}+\frac{3x^2}{x^3}+\frac{1}{x^3}}$$

$$= \frac{1+\frac{2}{x}+\frac{5}{x^3}}{5+\frac{3}{x}+\frac{1}{x^3}}$$

put $x\to\infty$

$$\frac{1+\frac{2}{\infty}+\frac{5}{\infty^3}}{5+\frac{3}{\infty}+\frac{1}{\infty^3}}$$

$$\frac{1+0+0}{5+0+0}$$

$$= \frac{1}{5}$$

(4) $\lim_{x\to\infty}\left(\frac{2x^4+3x^2-4x+9}{5x^4+2x^2-3x+7}\right)$

divide numerator & denumbrator by the highest power i.e x^4

$$= \frac{\frac{2x^4}{x^4}+\frac{3x^2}{x^4}+\frac{4x}{x^4}+\frac{9}{x^4}}{\frac{5x^4}{x^4}+\frac{2x^2}{x^4}-\frac{3x}{x^4}+\frac{7}{x^4}}$$

$$= \frac{2+\frac{3}{x^2}-\frac{4}{x^3}+\frac{9}{x^4}}{5+\frac{2}{x^2}-\frac{3}{x^3}+\frac{7}{x^4}}$$

put $x\to\infty$

$$\frac{2+\frac{3}{\infty^2}-\frac{4}{\infty^3}+\frac{9}{\infty^4}}{5+\frac{2}{\infty^2}-\frac{3}{\infty^3}+\frac{7}{\infty^3}}$$

$$\frac{2+0+0+0}{5+0+0+0}$$

$$= \frac{2}{5}$$

By Shortcut approach

$$\lim_{x\to\infty}\left(\frac{2x^4+3x^2-4x+9}{5x^4+2x^3-3x+7}\right)$$

Just take the coefficient of the highest power.
i.e
the highest power is x^4 and the coefficient is $\frac{2}{5}$
Hence, the answer is

$$\frac{2}{5}$$

(5) $$\lim_{x\to\infty}\left(\frac{x^2+3x+2}{x^2+4x+3}\right)$$

Just take the coefficient of the highest power i.e
the highest power is x^2 and the coefficient is

$$\frac{1}{1}=1$$

Hence, ans $= 1$

from (1)

$$\lim_{x\to\infty}\left(\frac{4x+3}{2x+1}\right)$$

do the same approach.
the highest power is x and the coefficient is $\frac{4}{2}=2$
Hence ans $= 2$

from (2)

$$\lim_{x\to\infty}\left(\frac{2x^2+3x+1}{4x^2+2x+2}\right)$$

the highest power is x^2 and the coefficient is $\frac{2}{4}=\frac{1}{2}$
Hence, ans $=\frac{1}{2}$

from (3)

$$\lim_{x\to\infty}\left(\frac{x^3+2x^2+5}{5x^3+3x^2+1}\right)$$

the highest power is x^3 and the coefficient is $\frac{1}{5}$
Hence, ans $=\frac{1}{5}$

(6) $\lim_{x\to\infty}\left(\frac{3x^3+4x^2+3x-5}{9x^3+8x^2+x-2}\right)$

the highest power is x^3 and the coefficient is $\frac{3}{9}$, Hence ans $= \frac{3}{9} = \frac{1}{3}$ //

(7) $\lim_{x\to\infty}\left(\frac{4x^3+3x^2-5}{8x^3+x+2}\right)$

the highest power is x^3 and the coefficient is $\frac{4}{8} = \frac{1}{2}$ Hence, ans $= \frac{1}{2}$ //

(8) $\lim_{x\to\infty}\left(\frac{2x^3+5x^2-6x+7}{3x^3+2x^2-7x+1}\right)$

the highest power is x^3 and coefficient is $\frac{2}{3}$, Hence ans $= \frac{2}{3}$ //

(9) $\lim_{x\to\infty}\left(\frac{7x^5-5x^3+6x^2+2x+3}{9x^5-10x^3+11x^2+4x+4}\right)$

the highest power x^5 and the coefficient is $\frac{7}{9}$, Hence, ans $= \frac{7}{9}$

(10) NOTE:
This method is only applicable if and Only if your $x\to\infty$.

Supposing your $x\to 0$ then you substitute direct e.g

$\lim_{x\to 0}\left(\frac{x^2+5x+9}{2x^2+3x+15}\right)$

put $x\to 0$

$\frac{(0)^2+5(0)+9}{2(0)^2+3(0)+15}$

$\frac{0+0+9}{0+0+15} = \frac{9}{15}$

$= \frac{3}{5}$ //

General revision on limit

(1) $\lim_{x \to 1} \left(\frac{x^2 - 1}{x - 1} \right)$

put $x \to 1$

$= \frac{(1)^2 - 1}{1 - 1} = \frac{1 - 1}{1 - 1} = \frac{0}{0}$

= undefined

Now, by L-hopital rule

$\lim_{x \to 1} \left(\frac{x^2 - 1}{x - 1} \right)$

$\frac{2x - 0}{1 - 0} = \frac{2x}{1}$

put $x \to 1$

$\frac{2(1)}{1} = 2$ //

Hence, ans = 2 //

(2) $\lim_{x \to 4} \left(7 - 2x + 5x^2 - 4x^3 \right)$

put $x \to 4$

$7 - 2(4) + 5(4)^2 - 4(4)^3$

(3) $\lim_{x \to -3} \left(\frac{x^2 - 9}{x + 3} \right)$

put $x \to -3$

$\frac{(-3)^2 - 9}{-3 + 3} = \frac{9 - 9}{-3 + 3} = \frac{0}{0}$

= undefined

Now,

By L-hospital rule

$\lim_{x \to -3} \left(\frac{x^2 - 9}{x + 3} \right)$

$\frac{2x - 0}{1 + 0} = \frac{2x}{1}$

put $x \to -3$

$\frac{2(-3)}{1} = \frac{-6}{1} = 6$

Hence, ans = -6 //

(4) $\lim_{x \to \pi} \left(x^2 - \sin x \right)$

recall that $\pi = 180^\circ$

put $x \to \pi$

$\lim_{x \to \pi} (x^2 - \sin x)$

$= (\pi)^2 - \sin(\pi)$

$= \pi^2 - \sin 180^\circ$

$\pi^2 - 0 = \pi^2$

because $\sin 180^\circ = 0$

Hence, ans = π^2

(5) $\lim_{x \to \pi} (x^2 \sin x)$

put $x \to \pi$

$\pi^2 \sin(\pi)$

$\pi^2 \sin(180^\circ)$

recall that $\sin 180 = 0$

$\pi^2 \times 0 = 0$

(6) $\lim_{x \to 0} \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)$

put $x \to 0$

$\frac{e^{(0)} - e^{(0)}}{e^{(0)} + e^{(0)}}$

but Note that anything raise power zero (0) is 1. excluding $0^0 = 0$

$= \frac{1-1}{1+1} = \frac{0}{2} = 0$

(7) $\lim_{x \to \infty} \left(\frac{2x^6 + 4x^5 + 6x^4 + 8x^3 + 12x^2 + 1}{7x^6 + 5x^5 + 3x^4 + 9x^3 + 6x^2 + 2} \right)$

the highest power is x^6

the coefficient of the highest is $\frac{2}{7}$

Hence ans = $\frac{2}{7}$

(8) $\lim_{x \to \infty} \left(\frac{1}{x^2} \right)$

put $x \to \infty$

$\frac{1}{\infty^2} = \frac{1}{\infty} = 0$

(9) $\lim_{x \to 1} \left(\frac{x^5+1}{x+1} \right)$

put $x \to 1$

$\frac{(1)^5+1}{1+1} = \frac{1+1}{1+1} = \frac{2}{2} = 1$ //

(10) $\lim_{x \to -1} \left(\frac{x^5-1}{x-1} \right)$

put $x \to -1$

$\frac{(-1)^5+1}{-1-1} = \frac{-1+1}{-1-1} = \frac{0}{0}$

= undefined

by L-hopital rule

$\frac{5x^4 - 0}{1-0}$

Put $x \to -1$

$\frac{5(-1)^4 - 0}{1-0} = 5$ //

Hence ans = 5 //

(11) $\lim_{x \to 1} \left(\cos(x^2-1) \right)$

put $x \to 1$

$\cos(1^2-1)$

$\cos(1-1)$

$\cos(0)$

but $\cos 0 = 1$

Hence, ans = 1 //

(12) $\lim_{x \to 0} \left(\frac{1}{x^4} \right)$

put $x \to 0$

$\frac{1}{0^4} = \frac{1}{0} = \infty$

Hence ans = ∞ //

Solved and compiled by Sheriff aka Schrödinger
08108145715 (35)

(13) $\lim_{\theta \to 0} \left(\frac{\sin 2\theta + \sin\theta}{\theta} \right)$

put $\theta \to 0$

$\frac{\sin(0) + \sin(0)}{0}$

but $\sin 0 = 0$

$= \frac{0+0}{0} = \frac{0}{0}$ = undefined

by L-hopital rule

$\lim \left(\frac{\sin 2\theta + \sin\theta}{\theta} \right)$

differentiating w.r.t θ

$\frac{2\cos 2\theta + \cos\theta}{1}$

put $\theta \to 0$

$\frac{2\cos(0) + \cos(0)}{1}$

but $\cos(0) = 1$

$\frac{2(1)+1}{1} = \frac{2+1}{1} = \frac{3}{1} = 3$

Hence ans = 3//

(14) $\lim_{\theta \to 0} \left(\frac{\sin 5\theta}{\sin 7\theta} \right)$

put $\theta \to 0$

$\frac{\sin\theta}{\sin\theta} = \frac{0}{0}$ = undefined

by L-hopital rule

$\frac{\sin 5\theta}{\sin 7\theta}$

differentiating w.r.t θ

$\frac{5\cos 5\theta}{7\cos 7\theta}$

put $\theta \to 0$

$\frac{5\cos(0)}{7\cos(0)}$ recall that $\cos 0 = 1$

$\frac{5 \times 1}{7 \times 1} = \frac{5}{7}$// $= \frac{5}{7}$//

Hence ans = $\frac{5}{7}$//

$5 \times 0 = 0$
$7 \times 0 = 0$

(15) $\lim_{t \to 0} \left(\frac{t^2+t}{t^2-t} \right)$

Put $t \to 0$

$\frac{0^2+0}{0^2-0} = \frac{0+0}{0-0} = \frac{0}{0}$ //

= undefined

by L-hopital rule

$\left(\frac{t^2+t}{t^2-1t} \right)$

$\frac{2t+1}{2t-1}$

Put $t \to 0$

$\frac{2(0)+1}{2(0)-1} = \frac{0+1}{0-1} = \frac{1}{-1}$

$= \frac{-1}{1} = -1$

Hence, ans $= -1$ //

(16) $\lim_{x \to 0} (x^2 + 9x + 1)$

Put $x \to 0$

$0^2 + 9(0) + 1$

$0 + 0 + 1$

$= 1$ //

(17) $\lim_{x \to 1} (x^3 - 5x^2 + 2x + 3)$

Put $x \to 1$

$(1)^3 - 5(1)^2 + 2(1) + 3$

$1 - 5 + 2 + 3$

$-4 + 2 + 3$

$-2 + 3 = 1$ //

(18) $\lim_{x \to 0} (x - 2x)$

$0 - 2(0) = 0 - 0 = 0$ //

(19) $\lim_{x \to 1} \left(\frac{1}{x} \right)$

$= \frac{1}{1} = 1$ //

(20) $\lim_{b \to 2} \left(\frac{1}{b^2} \right)$

$= \frac{1}{2^2} = \frac{1}{4}$ //

Solved and Compiled by Sheriff a.k.a Schrödinger
08108145725

DIFFERENTIATION

Consider this examples

① If $y = x^n$

$\frac{dy}{dx} = n \times x^{n-1}$

nx^{n-1}

This is the general formula for differentiation

② if $y = 2$

$\frac{dy}{dx} = 0$

Because differentiation of any Constant is zero(0)

③ If $y = 3$

$\frac{dy}{dx} = 0$//

④ If $y = 5$

$\frac{dy}{dx} = 0$//

Note: that differential of any constant is equal zero(0)
zero(0)

⑤ If $y = 100$

$\frac{dy}{dx} = 0$//

⑥ If $y = 500$

$\frac{dy}{dx} = 0$//

⑦ If $y = 2x^2$

$\frac{dy}{dx} = 2 \times 2x^{2-1}$

$4x$

⑧ If $y = 3x^2$

$\frac{dy}{dx} = 3 \times 2x^{2-1}$

$6x$

(9) if $y = 2x$

$\frac{dy}{dx} = 2 \times 1x^{1-1}$

$2x^{1-1} = 2x^0$

recall that anything raise to power zero (0) $= 1$

excluding $0^0 = 0$

So $2x^0 = 2 \times 1 = 2$ //

(10) if $y = x$.

$\frac{dy}{dx} = 1 \times x^{1-1}$

$x^{1-1} = x^0$

$\frac{dy}{dx} = 1$ //

(11) if $y = x^2$

$\frac{dy}{dx} = 2x$

(12) if $y = 6x^3$.

$\frac{dy}{dx} = 6 \times 3x^{3-1}$

$18x^2$

(13) if $y = 2x^4$

$\frac{dy}{dx} = 2 \times 4x^{4-1}$

$8x^3$

(14) if $y = (2x^2 + 1)$

$\frac{dy}{dx} = 2 \times 2x^{2-1} + 0$

$\frac{dy}{dx} = 4x$

(15) if $y = 5x^2 + 2x + 1$

$\frac{dy}{dx} = 5 \times 2x^{2-1} + 2 \times 1x^{1-1} + 0$

$10x + 2$

$e^x(\cos x + \sin x)$

Solved and compiled by Sheriff a.k.a Schrödinger

(16) If $y = 3x^4 - 5x^3 + x^2 + 2x - 4$

$\frac{dy}{dx} = 4x3x^{4-1} - 5x3x^{3-1} + 2x^{2-1} + 2x1x^{1-1} - 0$

$12x^3 - 15x^2 + 2x + 2$

(17) If $y = 4x^3 + x^2 - 6x - 3$

$\frac{dy}{dx} = 4x3x^{3-1} + 2x^{2-1} - 6x^{1-1} - 0$

$12x^2 + 2x - 6$

(18) If $y = x^3 - 2x^2 + 1$

$\frac{dy}{dx} = 3x^2 - 4x$

(19) If $y = ax$

$\frac{dy}{dx} = a$

(20) If $y = 2x^5 + 4x^4 - x^3 - 5x + 3x^2 + 7$

$\frac{dy}{dx} = 10x^4 + 16x^3 - 3x^2 - 5 + 6x + 0$

$= 10x^4 + 16x^3 - 3x^2 + 6x - 5$

(21) If $y = 3x^4 - 7x^3 + 4x^2 + 3x - 4$

find $\frac{dy}{dx}$ at $x = 2$

$\frac{dy}{dx} = 12x^3 - 21x^2 + 8x + 3 - 0$

$= 12x^3 - 21x^2 + 8x + 3$

at $x = 2$

Put $x = 2$

$= 12(2)^3 - 21(2)^2 + 8(2) + 3$

$12(8) - 21(4) + 16 + 3$

$96 - 84 + 16 + 3$

$96 - 84 + 19 = 31$

$\frac{dy}{dx} = 31$

(22) If $y = x^5 + 2x^4 - 4x^3 - 5x^2 + 8x - 3$

find $\frac{dy}{dx}$ at $x = -1$

$\frac{dy}{dx} = 5x^4 + 8x^3 - 12x^2 - 10x + 8$

Put $x = -1$

$5(-1)^4 + 8(-1)^3 - 12(-1)^2 - 10(-1) + 8$

$6(1)+8(1)-12(-1)^2-10(-1)+8$

$6-8-12+10+8$

$-3-12+10+8$

$-15+18=3$ //

(22) If $y=6x^3-7x^2+4x+5$

find $\frac{dy}{dx}$ at $x=3$

$\frac{dy}{dx}=18x^2-14x+4$

Put $x=3$

$18(3)^2-14(3)+4$

$18(9)-14(3)+4$

$162-42+4$

$=124$

(24) If $y=c$ find $\frac{dy}{dx}$

$\frac{dy}{dx}=0$ //

because C is not x and you are asked to find $\frac{dy}{dx}$ and recall that differential of any constant is zero (0)

(25) If $y=b$ find $\frac{dy}{dx}$?

$\frac{dy}{dx}=0$ //

(26) If $y=c$ find $\frac{dy}{dc}$

$\frac{dy}{dc}=1$ //

because here you are asked to find $\frac{dy}{dc}$ not $\frac{dy}{dx}$

(27) If $y=b$ find $\frac{dy}{db}$

$\frac{dy}{db}=1$ //

(28) If $y=ax^n$ find $\frac{dy}{dx}$?

$\frac{dy}{dx}=anx^{n-1}$

(29) If $y=bx$

$\frac{dy}{dx}=b$ //

(30) If $y=nx$

$\frac{dy}{dx}=n$ //

Differentiation by product rule

The product rule formular is given by

$$\frac{dy}{dx} = \frac{U dv + v dU}{dx} \quad \text{or} \quad \frac{dy}{dx} = U\frac{dv}{dx} + v\frac{dU}{dx}$$

Examples

① If $y = (x^2+5)(x^2-3)$ find y'

Note that $\frac{dy}{dx}$ is also known as y' i.e $\frac{dy}{dx} = y'$

let $U = x^2+5$

$V = x^2-3$

$\frac{dU}{dx} = 2x$

$\frac{dv}{dx} = 2x$

Substitute this into the formular i.e

$\frac{dy}{dx} = U\frac{dv}{dx} + v\frac{dU}{dx}$

$(x^2+5)2x + (x^2-3)(2x)$

by expanding

$2x^3 + 10x + 2x^3 - 6x$

$\frac{dy}{dx} = 4x^3 + 4x$

by Shortcut approach

Expand the bracket and differentiate it

$(x^2+5)(x^2-3)$

$x^4-3x^2+5x^2-15$

differentiate it

$4x^3-6x+10x-0$

$=4x^3+4x$

(2) If $y=(2x+1)(3x^2-5)$

let $u=2x+1$

$v=3x^2-5$

$\frac{du}{dx}=2$

$\frac{dv}{dx}=6x$

from the formular

$\frac{dy}{dx}=\frac{u\,dv}{dx}+\frac{v\,du}{dx}$

$\frac{dy}{dx}=(2x+1)(6x)+(3x^2-5)(2)$

$\frac{dy}{dx}=12x^2+6x+6x^2-10$

$18x^2+6x-10$

by shortcut

$(2x+1)(3x^2-5)$

(42)

expand the bracket and differentiate it

$6x^3-10x+3x^2-5$

$18x^2-10+6x-0$

$18x^2+6x-10$

(3) If $y=(x+1)(3x+2)$

let $u=x+1$

$v=3x+2$

$\frac{du}{dx}=1$

$\frac{dv}{dx}=3$

from the formular

$\frac{dy}{dx}=(x+1)(3)+(3x+2)(1)$

$3x+3+3x+2$

$6x+5=6x+5$

by shortcut

$(x+1)(3x+2)$

$3x^2+2x+3x+2$

$6x+2+3+0$

$6x+5$

Solved and compiled by Sheriff a.k.a Schrodinger
08108145725 (43)

(4) If $y=(x+5)(x+1)^2$

let $u=x+5$
let $v=(x+1)^2$
$\frac{du}{dx}=1$
$\frac{dv}{dx}=2\times1(x+1)$
$=2(x+1)$

$\frac{dy}{dx}=(x+5)(2x+2)+(x+1)^2\times1$
by expanding
$\frac{dy}{dx}=2x^2+10x+2x+10+(x+1)(x+1)$
$2x^2+10x+2x+10+x^2+2x+1$
$3x^2+14x+11$
$\frac{dy}{dx}=3x^2+14x+11$

(5) If $y=(x^2-x)(x+1)^2$

let $u=x^2-x$
let $v=(x+1)^2$
$\frac{du}{dx}=2x-1$
$\frac{dv}{dx}=2\times1(x+1)=(2x+2)$
$\frac{dy}{dx}=(x^2-x)(2x+2)+(x+1)^2(2x-1)$
by expanding
$2x^3-2x^2+2x^2-2x+(x^2+2x+1)(2x-1)$
$4x^3+6x^2+2x-3x^2-4x-1$
~~$2x^3-2x+2x^2-2x+2x^2+3x^2-1$~~
$4x^3+3x^2-2x-1$
$\frac{dy}{dx}=4x^3+3x^2-2x-1$

(6) If $y=x^3e^x$
$x^3e^x+e^x\cdot3x^2$
$x^3e^x+3x^2e^x$
$3x^2e^x+x^3e^x$

or
$x^3e^x+3x^2e^x$

y	derivative
e^x	e^x
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
x^n	nx^{n-1}
C	0
$\ln x$ or $\log x$	$\frac{1}{x}$
$\cot x$	$-\operatorname{cosec}^2 x$

(7) If $y \Rightarrow x^4 \cos x$

Consider this formular

$y' = uv' + vu'$

Implies

Skip first differentiate second plus skip second differentiate first

$x^4 \cos x$

$x^4(-\sin x) + \cos x(4x^3$

$-x^4 \sin x + 4x^3 \cos x$

$= 4x^3 \cos x - x^4 \sin x$

$4x^3 \cos x - x^4 \sin x$

or

$x^3(4\cos x - x\sin x)$

(8) If $y \Rightarrow x^5 e^x$

$x^5 e^x + e^x 5x^4$

$5x^4 e^x + x^5 e^x$

(9) $x^6 \cos x$

$x^6(-\sin x) + \cos x(6x^5)$

$-x^6 \sin x + 6x^5 \cos x$

$6x^5 \cos x - x^6 \sin x$

or

$x^5(6\cos x - x\sin x)$

Solved and compiled by Sheriff a.k.a Schudinger
08108145125 (45)

(10) If $y = 2x^5 \cos x$

$2x^5(-\sin x) + \cos x(10x^4)$

$-2x^5 \sin x + 10x^4 \cos x$

$10x^4 \cos x - 2x^5 \sin x$

or

$x^4(10\cos x - 2x\sin x)$

(11) If $y = 4x^3 \sin x$

$4x^3(\cos x + \sin x\, 12x^2$

$4x^3 \cos x + 12x^2 \sin x$

(12) If $y = \cos x \sin x$

$\cos x(\cos x) + \sin x(-\sin x)$

$\cos^2 x - \sin^2 x$

(13) If $y = e^x \sin x$

$e^x \cos x + \sin x\, e^x$

$e^x \cos x + e^x \sin x$

$e^x(\cos x + \sin x)$

(14) If $y = e^{5x}(3x+1)$

$e^{5x}(3) + (3x+1)5e^{5x}$

$3e^{5x} + 5e^{5x}(3x+1)$

$e^{5x}(3 + 5(3x+1)$

$e^{5x}(3 + 15x + 5)$

$e^{5x}(8 + 15x)$

(15) If $y = e^{2x} \ln 5x$

$e^{2x}\left(\frac{5}{5x}\right) + \ln 5x(2e^{2x})$

$e^{2x}\left(\frac{\cancel{5}}{\cancel{5}x} + 2\ln 5x\right)$

$e^{2x}\left(\frac{1}{x} + 2\ln 5x\right)$

$e^{2x}\left(\frac{1}{x} + 2\ln 5x\right)$

(16)

(16) If $y = x^2 \tan x$

$x^2 \sec^2 x + \tan x \, 2x$

$x^2 \sec^2 x + 2x \tan x$

(17) If $y = x^3 \sin 5x$

$x^3 (5\cos 5x) + \sin 5x (3x^2)$

$x^3 5\cos 5x + 3x^2 \sin 5x$

$5x^3 \cos 5x + 3x^2 \sin 5x$

or

$x^2 (5x \cos 5x + 3\sin 5x)$

(18) If $y = x \cos 2x$

$x(-2\sin 2x) + \cos 2x (1)$

$-2x \sin 2x + \cos 2x$

$\cos 2x - 2x \sin 2x$

(19) If $y = x^3 \sin 7x$

$x^3 7\cos 7x + \sin 7x (3x^2)$

$7x^3 \cos 7x + 3x^2 \sin 7x$

$x^2 (7x \cos 7x + 3\sin 7x)$

(20) $x^4 \tan 2x$

$x^4 (2\sec^2 2x) + \tan 2x (4x^3)$

$2x^4 \sec^2 2x + 4x^3 \tan 2x$

or

$2x^2 (x^2 \sec^2 2x + 2x \tan 2x)$

$2x^2 (x^2 \sec^2 2x + 2x \tan 2x)$

Differentiation by quotient rule (47)

The quotient rule formular is given by

$$\frac{dy}{dx} = \frac{v\,du - u\,dv}{v^2}$$

Examples

① If $y = \frac{x^2}{x+2}$ → u / → v

Let $u = x^2$

Let $v = x+2$

$\frac{du}{dx} = 2x$

$\frac{dv}{dx} = 1$

from the formular

$$\frac{(x+2)2x - x^2(1)}{(x+2)^2}$$

$$\frac{2x^2+4x-x^2}{(x+2)^2}$$

$$\frac{x^2+4x}{(x+2)^2}$$

② If $y = \frac{1+x^2}{1+x^2}$ — u / — v

Let $u = 1+x^2$

$v = 1+x^2$

$\frac{du}{dx} = 2x$

$\frac{dv}{dx} = +2x$

$$\frac{(1-x^2)(2x) - (1+x^2)(+2x)}{(1+x^2)^2}$$

$$\frac{2x - 2x^3 - 2x - 2x^3}{(1+x^2)^2}$$

$$\frac{-4x^3}{(1+x^2)^2}$$

(3) If $y = \frac{x+2}{2x+1}$

Let $u = x+2$ $\quad v = (2x+1)^2$

$v = 2x+1$,

$\frac{du}{dx} = 1$

$\frac{dv}{dx} = 2$

$\frac{(2x+1)(1) - (x+2)(2)}{(2x+1)^2}$

$\frac{2x+1-2x-4}{(2x+1)^2}$

$\frac{-3}{(2x+1)^2} = \frac{-3}{(2x+1)^2}$

(4)

If $y = \frac{\sin 3x}{x+1}$

Let $u = \sin 3x$

$v = x+1$

$\frac{du}{dx} = 3\cos 3x$

$\frac{dv}{dx} = 1$,

$\frac{(x+1)(3\cos 3x) - \sin 3x(1)}{(x+1)^2}$

$\frac{(x+1)3\cos 3x - \sin 3x}{(x+1)^2}$

(5)

If $y = \frac{3x^4 - x}{x^3}$ $\quad u = 3x^4 - x$

$v = x^3$

$\frac{du}{dx} = 12x^3 - 1$

$\frac{dv}{dx} = 3x^2$

$\frac{x^3(12x^3-1) - (3x^4-x)(3x^2)}{(x^3)^2}$

$\frac{12x^6 - x^3 - 9x^6 - 3x^3}{(x^3)^2}$

$\frac{3x^6 - 4x^3}{x^6} = x^3\left(\frac{3x^3-4}{x^3}\right)$

$= \frac{3x^3-4}{x^3}$

Solved and compiled by Sheriff aka Schrodinger
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(49)

⑥ If $y = \frac{\cos 2x}{x^2}$

Let $u = \cos 2x$
$v = x^2$

$\frac{du}{dx} = -2\sin 2x$

$\frac{dv}{dx} = 2x$

$\frac{-x^2(2\sin 2x) - \cos 2x(2x)}{(x^2)^2}$

$\frac{-2x^2 \sin 2x - 2x\cos 2x}{x^4}$

factorize x

$\frac{x(-2x\sin 2x - 2\cos 2x)}{x \times x^3}$

$\frac{2x\sin 2x - 2\cos 2x}{x^3}$

$\frac{-2(x\sin 2x + \cos 2x)}{x^3}$

⑦ If $y = \frac{\sin x}{\cos x}$

$u = \sin x$
$v = \cos x$

$\frac{du}{dx} = \cos x$
$\frac{dv}{dx} = -\sin x$

$\frac{\cos x(\cos x) - \sin x(-\sin x)}{(\cos x)^2}$

$\frac{\cos^2 x + \sin^2 x}{(\cos x)^2}$

recall from trigonometry Identity or from your secondary school that

$\cos^2 x + \sin^2 x = 1$

$\frac{\cos^2 x + \sin^2 x}{(\cos x)^2} = \frac{1}{\cos^2 x}$

or $= \sec^2 x$

because $\sec x = \frac{1}{\cos x} \Rightarrow \sec^2 x = \frac{1}{\cos^2 x}$

(8) If $y = \frac{x^2-2}{x^2+3}$

Let $u = x^2-2$

$v = x^2+3$

$\frac{du}{dx} = 2$

$\frac{dv}{dx} = 2$

$\frac{(x^2+3)(2)-(x^2-2)(2)}{(x^2+3)^2}$

$\frac{2x^2+6-2x^2+4}{(x^2+3)^2}$

$\frac{dy}{dx} = \frac{10}{(x^2+3)^2}$

(9) If $y = \frac{1-2x}{x+2}$

$u = 1-2x \quad \frac{du}{dx} = -2$

$v = x+2 \quad \frac{dv}{dx} = 1$

$\frac{dy}{dx} = \frac{(x+2)(-2)-(1-2x)1)}{(x+2)^2}$

$\frac{-2x-4-1+2x}{(x+2)^2}$

$= \frac{-5}{(x+2)^2}$

(10) $\frac{1-4}{x^3-4} \quad \frac{du}{dx} = 0$

$\frac{dv}{dx} = 3x^2$

$\frac{dy}{dx} = \frac{x^3(0)-1(3x^2)}{(x^3)^2}$

$0 - \frac{3x^2}{x^6} =$

$= \frac{-3x^2}{x6}$

$= \frac{-3}{x^3}$

Solved and compiled by Sheriff 4Dg Schrödinger
08057485725

Differentiation by chain rule or function of function

If y is a function of u and u is a function of x. Then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Examples

(1) $y = (x^2 - 2x + 1)^3$

let $u = x^2 - 2x + 1$

let $y = u^3$

$\frac{du}{dx} = 2x - 2$

$\frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = 3u^2 \times 2x - 2$

but recall that

$u = x^2 - 2x + 1$

$= 3(x^2 - 2x + 1)^2 (2x - 2)$

(2) $y = \frac{(6x-6)(x^2-2x+1)^2}{(x^2+2)^4}$ or

let $u = x^2 + 2$

$\frac{du}{dx} = 2x$

let $y = u^4$

$\frac{dy}{du} = 4u^3$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$4u^3 \times 2x$

but recall that

$u = x^2 + 2$

$4(x^2+2)^3 \times 2x$

$8x(x^2+2)^3$

(3) $y = (x^2 + 1)^3$

let $u = x^2 + 1$

$\frac{du}{dx} = 2x$

let $y = u^3$

$\frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = \frac{dy}{dv} \times \frac{dv}{dx}$

$= 3v^3 \times 2x$

but recall that $v = x^2+1$

$3(x^2+1)^3 \times 2x$

$6x(x^2+1)^2$

(4) $y = (3x+2)^{-2}$

let $v = 3x+2$

$\frac{dv}{dx} = 3$

let $y = v^{-2}$

$\frac{dy}{dv} = -2v^{-2-1}$

$= -2v^{-3}$

$\frac{dy}{dx} = \frac{dy}{dv} \times \frac{dv}{dx}$

$-2v^{-3} \times 3$

but $v = (3x+2)$

$-2(3x+2)^{-3} \times 3$

$-6(3x+2)^{-3}$

by shortcut approach

from (1)

$(x^2-2x+1)^3$

drop the power then multiply it by differentiate of the function in the bracket also rewrite the whole function i.e $(x^2-2x+1)^3$ then minus 1 from the power

i-e

$3x(2x-2)(x^2-2x+1)^2$

$3(x^2-2x+1)^2(2x-2)$

or

$(6x-6)(x^2-2x+1)^2$

Solved and compiled by Sheriff aka Schrodinger 08108145725 (53)

Shortcut approach

from (2)

$y=(x^2+1)^4$

drop the power multiply it by the differential of the function in the bracket also rewrite the whole function i.e $(x^2+1)^4$ then minus 1 from the power

i.e $= 4 \times 2x(x^2+1)^{4-1}$

$8x(x^2+1)^3$

from (3)

$y=(x^2+1)^3$

drop the power (3) multiply it by the differential of the function in the bracket $2x$, also rewrite the whole function i.e $(x^2+1)^3$ then minus 1 from the power

$= 3 \times 2x(x^2+1)^{3-1}$

$6x(x^2+1)^2$

(5) $y=(2x^2-4x)^5$

let $u = 2x^2-4x$

$\frac{du}{dx} = 4x-4$

let $y = u^5 = u^5$

$\frac{dy}{du} = 5u^4$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = 5u^4 \times 4x-4$

but $u = 2x^2-4x$

$5(2x^2-4x)^4(4x-4)$

$5(2x^2-4x)^4(4x-4)$

Shortcut approach

$5x(4x-4)(2x^2-4x)^{5-1}$

$5(2x^2-4x)^4(4x-4)$

(6) $y=(4x-5)^6$

let $u=4x-5$

$\frac{du}{dx}=4$

let $y=u^6$

$\frac{dy}{du}=6u^5$

$\frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}$

$6u^5\times 4$

but $u=4x-5$

$6(4x-5)^5(4)$

$24(4x-5)^5$

by shortcut

$6\times 4(4x-5)^{6-1}$

$24(4x-5)^5$

(7) $y=\sqrt{x^2-2}\Rightarrow(x^2-2)^{1/2}$

let $u=x^2-2$

$\frac{du}{dx}=2x$

let $y=u^{1/2}$

$\frac{dy}{du}=\frac{1}{2}u^{\frac{1}{2}-1}=\frac{1}{2}u^{-1/2}$

$\frac{dy}{du}=\frac{1}{2u^{1/2}}\Rightarrow\frac{1}{2\sqrt{u}}$

$\frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}$

$\frac{dy}{dx}=\frac{1}{2\sqrt{u}}\times 2x$

$\frac{2x}{2\sqrt{u}}$

but $u=x^2-2$

$\frac{2x}{2\sqrt{x^2-2}}=\frac{x}{\sqrt{x^2-2}}$

(8) $y=(2x+5)^{1/2}$

let $u=2x+5$

$\frac{du}{dx}=2$

let $y = u^{1/2}$

$\frac{dy}{du} = \frac{1}{2\sqrt{u}}$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{2\sqrt{u}} \times 2$

$\frac{2}{2\sqrt{u}}$

but $u = 2x+5$

$\frac{2}{2\sqrt{2x+5}} = \frac{\cancel{2} \times 1}{\cancel{2}\sqrt{2x+5}}$

$= \frac{1}{\sqrt{2x+5}}$

(9) $y = \sqrt{(x^2+6x-1)}$

$= (x^2+6x-1)^{1/2}$

let $u = x^2+6x-1$

$\frac{dv}{dx} = 2x+6$

let $y = u^{1/2}$

$\frac{dy}{du} = \frac{1}{2\sqrt{u}}$

$\frac{dy}{dx} = \frac{1}{2\sqrt{u}} \times 2x+6$

$\frac{2x+6}{2\sqrt{u}}$

but $u = x^2+6x-1$

$\frac{2x+6}{2\sqrt{x^2+6x-1}} = \frac{\cancel{2}x(x+3)}{\cancel{2}x\sqrt{x^2+6x-1}}$

$= \frac{x+3}{\sqrt{x^2+6x-1}}$

(10) $y = 3\sqrt{x^3-3x+2}$

$= (x^3-3x+2)^{1/3}$

let $u = x^3-3x+2$

$\frac{du}{dx} = 3x^2-3$

let $y = u^{1/3}$

$\frac{dy}{du} = \frac{1}{3} u^{\frac{1}{3}-1}$

$= \frac{1}{3} u^{-\frac{2}{3}}$

$\frac{dy}{du} = \frac{1}{3} u^{\frac{2}{3}} \Rightarrow \frac{1}{3(\sqrt[3]{u^2})}$

$\frac{dy}{dx} = \frac{1}{3(\sqrt[3]{u^2}} \times 3x^2 - 3$

$\frac{3x^2 - 3}{3(\sqrt[3]{u^2}}$

but

$u = x^3 - 3x + 2$

$\frac{3x^2 - 3}{3\left(\sqrt[3]{x^3 - 3x + 2}\right)^2}$

factor. 2e 3

$\frac{3 \times (x^2 - 1)}{3\left(\sqrt[3]{x^3 - 3x + 2}\right)^2}$

$\frac{x^2 - 1}{\left(\sqrt[3]{x^3 - 3x + 2}\right)^2}$

$\frac{dy}{dx} = \frac{x^2 - 1}{3\left(\sqrt[3]{x^3 - 3x + 2}\right)^2}$

$\frac{dy}{dx} = \frac{x^2 - 1}{\left(3\sqrt[3]{x^3 - 3x + 2}\right)^2}$

by short cut

$\sqrt[3]{x^3 - 3x + 2} = (x^3 - 3x + 2)^{1/3}$

$\frac{1}{3} \times 3x^2 - 3\left(x^3 - 3x + 2\right)^{\frac{1}{3}-1}$

$\frac{3x^2 - 3}{3}\left(x^3 - 3x + 2\right)^{-\frac{2}{3}}$

$= \frac{3(x^2 - 1)}{3(1)}\left(x^3 - 3x + 2\right)^{-\frac{2}{3}}$

$\frac{x^2 - 1}{3\sqrt[3]{x^3 - 3x + 2}^2}$

(11)

$y = \cos^3(3x) = (\cos 3x)^3$

let $u = \cos 3x$

$\frac{du}{dx} = -3\sin 3x$

let $y = u^3$

$\frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = 3u^2 \times -3\sin 3x$

$-9u^2 \sin 3x$

but $u = \cos 3x$

$-9(\cos 3x)^2 \sin 3x$

$-9\cos^2 3x \sin 3x$

Or

$-9\sin 3x \cos^2 3x$

(12)

$y = \sin^2 x = (\sin x)^2$

let $u = \sin x$

$\frac{du}{dx} = \cos x$

let $y = u^2$

$\frac{dy}{du} = 2u$

$\frac{dy}{dx} = 2u \times \cos x$

but $u = \sin x$

$2\sin x \cos x$

$2\sin x \cos x$

but $2\sin x \cos x$ from trig identity $= \sin 2x$

$\Rightarrow 2\sin x \cos x = \sin 2x$

(13)

$y = (x^3 - 2)^7$

$= 7 \times 3x^2 (x^3 - 2)^{7-1}$

$21x^2(x^3 - 2)^6$

(14) $y = (5x^2 - 1)^2$

$2 \times 10x(5x^2 - 1)^{2-1}$

$10x(5x^2 - 1)$

(15) $y = \sqrt[4]{x^3 - 3x^2 - 6x - 1}$

$= (x^3 - 3x^2 - 6x - 1)^{1/4}$

let $u = x^3 - 3x^2 - 6x - 1$

$\frac{du}{dx} = 3x^2 - 6x - 6$

let $y = u^{1/4}$

$\frac{dy}{du} = \frac{1}{4} u^{\frac{1}{4}-1}$

$\Rightarrow \frac{1}{4} u^{-3/4} = \frac{1}{4u^{3/4}}$

$\frac{dy}{du} = \frac{1}{4(\sqrt[4]{u^3})}$

$\frac{dy}{dx} = \frac{1}{4(\sqrt[4]{u^3})} \times 3x^2 - 6x - 6$

$\frac{3x^2 - 6x - 6}{4(\sqrt[4]{u^3})}$

but

$u = x^3 - 3x^2 - 6x - 1$

$\frac{dy}{dx} = \frac{3x^2 - 6x - 6}{4(\sqrt[4]{x^3 - 3x^2 - 6x - 1})^3}$

(16) $y = (1 - 2x)^9$

$9 \times 2 (1 - 2x)^{9-1}$

$18(1 - 2x)^8$

(17) $y = (9x^2 - 1)^8$

$8 \times 18x (9x^2 - 1)^{8-1}$

$144x (9x^2 - 1)^7$

(18) $y = (1 - nx)^4$

$4 \times -n (1 - nx)^{4-1}$

$-4n(1 - nx)^3$

Solved and compiled by Sheriff at a Schrödinger (51)

Differentiation by Implicity function

This is a function in which the dependent variable and independent variable are not defined clearly.

Consider the shortcut below

formular

$$\frac{dy}{dx} = -\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

$\frac{\partial f}{\partial x}$ → differentiate partially respect to x

$\frac{\partial f}{\partial y}$ → differentiate partially respect to y

Examples

$y = x^2 + y^2 = 2$

$x^2 + y^2 - 2 = 0$

$$-\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = -\left(\frac{2x + 0 - 0}{0 + 2y - 0}\right)$$

$$\frac{dy}{dx} = \frac{-2x}{2y} = -\frac{x}{y}$$

② $y = x^3 + y^3 = 5$

$x^3 + y^3 - 5 = 0$

$$-\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = -\left(\frac{3x^2 + 0 - 0}{0 + 3y^2}\right)$$

$$= \frac{-3x^2}{3y^2} = -\frac{x^2}{y^2}$$

$$\frac{dy}{dx} = -\frac{x^2}{y^2}$$

③ $y = x^2 + 2y + 2x^3 = 5$

$x^2 + 2y + 2x^3 - 5 = 0$

$$-\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = -\left(\frac{2x + 0 + 6x^2 - 0}{0 + 2 + 0 - 0}\right)$$

$$= -\left(\frac{2x + 6x^2}{2}\right)$$

$$-2\left(\frac{x + 3x^2}{2}\right)$$

$$2\left(\frac{1}{1}\right)$$

$$-\left(\frac{x + 3x^2}{1}\right)$$

$$= -(x + 3x^2)$$

(4) If $y = x^3 + 2xy + 3xy^2 = 10$

$f = x^2 + 2xy + 3xy^2 - 10 = 0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right) = -\left(\frac{2x + 2y + 3y^2 - 0}{0 + 2x + 6xy - 0}\right)$$

$$= -\left(\frac{2x + 2y + 3y^2}{2x + 6xy}\right)$$

(5) $y = x^3 + y^3 + 3xy^2 = 8$

$= x^3 + y^3 + 3xy^2 - 8 = 0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right) = -\left(\frac{3x^2 + 0 + 3y^2 = 0}{0 + 3y^2 + 6xy - 0}\right)$$

$$= -\left(\frac{3x^2 + 3y^2}{3y^2 + 6xy}\right)$$

$$\frac{3}{3}-\left(\frac{x^2 + y^2}{y^2 + 2xy}\right)$$

$$= -\left(\frac{x^2 + y^2}{y^2 + 2xy}\right)$$

(60)

(6) $y = xy = 8$

$= xy - 8 = 0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right) = -\left(\frac{y - 0}{x - 0}\right)$$

$$= -y/x$$

(7) $y = (x+y)^2 = 6x^2y$

Expand $(x+y)^2$

$(x+y)(x+y)$

$x^2 + xy + xy + y^2$

$x^2 + 2xy + y^2$

$= x^2 + 2xy + y^2 - 6x^2y = 0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right) = -\left(\frac{2x + 2y + 0 - 12xy}{0 + 2x + 2y - 6x^2}\right)$$

$$= -\left(\frac{2x + 2y - 12xy}{2x + 2y - 6x^2}\right)$$

by 2

$$= -\left(\frac{x + y - 6xy}{x + y - 3x^2}\right)$$

Solved and Compiled by Sheriff aka Schrödinger

05108145725

(8) $x^3+y^2+3x^2y+4xy^3=6$

$x^3+y^2+3x^2y+4xy^3-6=0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right)=-\left(\frac{3x^2+0+6xy+4y^3-0}{0+2y+3x^2+12xy^2-0}\right)$$

$$=-\left(\frac{3x^2+6xy+4y^3}{2y+3x^2+12xy^2}\right)$$

(9) $x^5+y^5+4x^3y=5xy^2$

$x^5+y^5+4x^3y-5xy^2=0$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right)=-\left(\frac{5x^4+0+12x^2y-5y^2}{0+5y^4+4x^3-10xy}\right)$$

$$=-\left(\frac{5x^4+12x^2y-5y^2}{5y^4+4x^3-10xy}\right)$$

(10) $y=x^2+y^2$ find $\frac{dy}{dx}$ at $x=3, y=2$

$$-\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right)=-\left(\frac{2x+0}{0+2y}\right)=-\frac{2x}{2y}=-\frac{x}{y}$$

put value of x and y

$$\frac{dy}{dx}=\frac{-x}{y}=-\frac{3}{2}$$

$$=-3/2//$$

(11) $y=2xy+\sin x+\tan y$

$$-\left(\frac{2y+\cos x+0}{2x+0+\sec^2 y}\right)$$

$$=-\left(\frac{2y+\cos x}{2x+\sec^2 y}\right)$$

(12) $y=x^2+y^2-2x-6y+5=0+3$

find y' at $x=3, y=2$

$$=-\left(\frac{2x+0-2-0+0}{0+2y-0-6+0}\right)$$

$$=-\left(\frac{2x-2}{2y-6}\right)$$

Put $x=3, y=2$

$$=-\left(\frac{2(3)-2}{2(2)-6}\right)$$

$$-\left(\frac{6-2}{4-6}\right)=-\frac{4}{-2}$$

$$=+\left(\frac{4}{2}\right)=2//$$

(13) $x^4 + y^4 + 6x^2y^2 + 4xy + 5x^3y = 6x^2y \Rightarrow x^4 + y^4 + 6x^2y^2 + 4xy + 5x^3y - 6x^2y = 0$

$$= -\left(\frac{4x^3 + 0 + 12xy^2 + 4y + 15x^2y - 12xy}{0 + 4y^3 + 12x^2y + 4x + 5x^3 - 6x^2}\right)$$

$$= -\left(\frac{4x^3 + 12xy^2 + 4y + 15x^2y - 12xy}{4y^3 + 12x^2y + 4x + 5x^3 - 6x^2}\right)$$

(14) $(x+y)^3 = 4x^2y$

Expand $(x+y)^3$ using binomial expansion

1
1 1
1 2 1
1 3 3 1

$x^3y^0 + x^2y^1 + x^1y^2 + x^0y^3$

$x^3 + x^2y + xy^2 + y^3$

$x^3 + 3x^2y + 3xy^2 + y^3$

$= x^3 + 3x^2y + 3xy^2 + y^3 - 4x^2y = 0$

$$= -\left(\frac{3x^2 + 6xy + 3y^2 - 8xy}{3x^2 + 6xy + 3y^2 - 4x^2}\right)$$

Differentiation by logarithmic functions

Consider the following terms

If $y = \log_a x$ ——— *

$\frac{dy}{dx} = \frac{1}{x}\log_e a$

If $y = \log_e x$ ——— **

$\frac{dy}{dx} = \frac{1}{x}\log_e e = \frac{1}{x}$

$\frac{dy}{dx} = \frac{1}{x}$

Note that log and ln are the same

$\boxed{\log = \ln}$

Examples

① If $y = \log_a(x^2+3)$

Let $u = x^2+3$

$\frac{du}{dx} = 2x$

Let $y = \log_a u$

$\frac{dy}{du} = \frac{1}{u}\log_e a$

by chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{u}\log_e a \times 2x$

$= \frac{2x}{u}\log_e a$

but $u = x^2+3$

$\frac{dy}{dx} = \frac{2x}{x^2+3}\log_e a$

② $y = \log_a(x^3-1)$

Let $u = x^3-1$

$\frac{du}{dx} = 3x^2$

Let $y = \log_a u$

$\frac{dy}{du} = \frac{1}{u}\log_e a$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \Rightarrow \frac{1}{u}\log_e a \times 3x^2$

$= \frac{3x^2}{u}\log_e a$

$u = x^3-1$

$= \frac{3x^2}{x^3-1}\log_e a$

③ $y = \log_e(x^2-4)$

let $u = x^2-4$

$\frac{du}{dx} = 2x$

let $y = \log_e u$

$\frac{dy}{du} = \frac{1}{u}\log_e e = \frac{1}{u}$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{u} \times 2x = \frac{2x}{u}$

put $u = x^2-4$

$= \frac{2x}{x^2-4}$

④ $y = \ln \cos 3x$

recall that $\ln = \log$

let $u = \cos 3x$

$\frac{du}{dx} = -3\sin 3x$

let $y = \ln u$

$\frac{dy}{du} = \frac{1}{u}$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{u} \times -3\sin 3x$

$= \frac{-3\sin 3x}{u}$

Put $u = \cos 3x$

$= \frac{-3\sin 3x}{\cos 3x}$

recall that

$\tan x = \frac{\sin x}{\cos x}$

$= \tan 3x = \frac{\sin 3x}{\cos 3x}$

$= -3\tan 3x$

⑤ $\ln(3-4\cos x)$

let $u = 3-4\cos x$

$\frac{du}{dx} = 0 - -4\sin x$

$= 4\sin x$

let $y = \ln u$

$\frac{dy}{du} = \frac{1}{u}$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{u} \times 4\sin x$

$\frac{4\sin x}{u}$

Put $u = 3 - 4\cos x$

$= \frac{4\sin x}{3 - 4\cos x}$

(6) $\log_{10}(2x-1)$

Let $u = 2x - 1$

$\frac{du}{dx} = 2$

Let $y = \log_{10} u$

$\frac{dy}{du} = \frac{1}{u}\log_{10}$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{u}\log_{10} \times 2$

$\frac{2}{u}\log_{10}$

put $u = 2x - 1$

$\frac{2}{2x-1}\log_{10}$ or $\frac{2}{2x-1}\ln 10$

(7) $\log_a x$

Let $u = x$ $\frac{du}{dx} = 1$

Let $y = \log_a u$

$\frac{dy}{du} = \frac{1}{u}\log a$

$\frac{dy}{dx} = \frac{1}{u}\log a \times 1$

$= \frac{1}{u}\log a$

put $u = x$

$= \frac{1}{x}\log a$ or $\frac{1}{x}\ln a$

(8) $\ln 4x$

Let $u = 4x$ $\frac{du}{dx} = 4$

Let $y = \ln u$ $\frac{dy}{du} = \frac{1}{u}$

$\frac{dy}{dx} = \frac{1}{u} \times 4 = \frac{4}{u}$

Put $u = 4x$

$\frac{4}{4x} = \frac{1}{x}$ //

Differentiation by exponential function (66)

① If $y = e^x$ $\frac{dy}{dx} = e^x$

② If $y = e^{3x}$

let $u = 3x$

$\frac{du}{dx} = 3$

let $y = e^u$

$\frac{dy}{du} = e^u$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = e^u \times 3x = 3e^u$

Put $u = 3x$

$\frac{dy}{dx} = 3e^{3x}$

Shortcut approach

$y = e^{3x}$

Just differentiate the power i.e $3x$ then multiply it by the whole function that is

$y = e^{3x}$

$3x\, e^{3x} = 3e^{3x}$

③ $y = e^{4x}$

let $u = 4x$ $\frac{du}{dx} = 4$

let $y = e^u$

$\frac{dy}{du} = e^u$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = e^u \times 4 =$

$4e^u$

Put $u = 4x$

$= 4e^{4x}$

Shortcut

$y = e^{4x}$

$= 4x\, e^{4x}$

$= 4e^{4x}$

(4)

$y = e^{-6x}$

let $u = -6x$

$\frac{du}{dx} = -6$

let $y = e^{-u} = e^{u}$

$\frac{dy}{du} = e^{u}$

$\frac{dy}{dx} = e^{u} \times -6$

$= -6e^{u}$

put $u = -6x$

$= -6e^{-6x}$

(5)

$y = e^{3x+1}$

let $u = 3x+1$

$\frac{du}{dx} = 3$

let $y = e^{u}$

$\frac{dy}{du} = e^{u}$

$\frac{dy}{dx} = e^{u} \times 3$

$3e^{u}$

put $u = 3x+1$

$= 3e^{3x+1}$

(6)

$y = e^{3x^2+2x-1}$

let $u = 3x^2+2x-1$

$\frac{du}{dx} = 6x-2$

let $y = e^{u}$

$\frac{dy}{du} = e^{u}$

$\frac{dy}{dx} = e^{u} \times 6x-2$

$6x-2 \times e^{u}$

$(6x-2)e^{u}$

put $u = 3x^2+2x+1$

$(6x-2)e^{3x^2+2x-1}$

(7) $y = e^{\sin x}$

let $u = \sin x$

$\frac{du}{dx} = \cos x$

let $y = e^u$

$\frac{dy}{du} = e^u$

$\frac{dy}{dx} = e^u \times \cos x$

$\cos x\, e^u$

put $u = \sin x$

$\cos x\, e^{\sin x}$

(8) $x^2 e^{3x}$

$x^2(3e^{3x}) + e^{3x}(2x)$

$3e^{3x}x^2 + 2xe^{3x}$

$3x^2e^{3x} + 2xe^{3x}$

(9) $y = e^{\sin 2x}$

let $u = \sin 2x$

$\frac{du}{dx} = 2\cos 2x$

let $y = e^u$

$\frac{dy}{du} = e^u$

$\frac{dy}{dx} = e^u \times 2\cos 2x$

$2\cos 2x\, e^u$

put $u = \sin 2x$

$2\cos 2x\, e^{\sin 2x}$

(10) $y = e^{\tan x}$

let $u = \tan x$

$\frac{du}{dx} = \sec^2 x$

let $y = e^u$

$\frac{dy}{du} = e^u$

$\frac{dy}{dx} = e^u \times \sec^2 x$

$\sec^2 x\, e^{\tan x}$

Solved and Compiled by Sheriff aka Schrödinger

Differentiation by trigonometry function

$$\frac{\sin\theta}{\cos\theta} = \tan\theta$$

$$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\sec x$	$\sec x \tan x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sin ax$	$a\cos ax$
$\cos ax$	$-a\sin ax$
$\tan ax$	$a\sec^2 ax$

Examples

① $y = \sin 5x$

let $u = 5x$

$\frac{du}{dx} = 5$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \Rightarrow \cos u \times 5$

put $u = 5x$

$5\cos 5x$

② $y = \sin 3x$

let $u = 3x$

$\frac{du}{dx} = 3$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \cos u \times 3$

$3\cos u$

Put $u = 3x$

$3\cos 3x$

(3) $y = \cos^2 x = (\cos x)^2$

let $u = \cos x$

$\frac{du}{dx} = -\sin x$

let $y = u^2$

$\frac{dy}{du} = 2u$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = 2u \times -\sin x$

$-2u \sin x$

put $u = \cos x$

$-2 \cos x \sin x =$

from trig identity

$-2 \cos x \sin x = -\sin 2x$

(4) $\sin^3 x = (\sin x)^2$

let $u = \sin x$

$\frac{du}{dx} = \cos x$

let $y = u^2$

$\frac{dy}{du} = 2u$

$\frac{dy}{dx} = 2u \times \cos x$

put $u = \sin x$

$2 \sin x \cos x = \sin 2x$

(5) $\tan^5 x = (\tan x)^5$

let $u = \tan x$

$\frac{du}{dx} = \sec^2 x$

let $y = u^5$

$\frac{dy}{du} = 5u^4$

$\frac{dy}{dx} = 5u^4 \times \sec^2 x$

$5u^4 \sec^2 x$

put $u = \tan x$

$5 \tan^4 x \sec^2 x$

(6) $\tan 4x$

let $u = 4x$

$\frac{du}{dx} = 4$

let $y = \tan u$

$\frac{dy}{du} = \sec^2 u$

$\frac{dy}{dx} = \sec^2 u \times 4$

$= 4\sec^2 u$

put $u = 4x$

$4\sec^2 4x$

⑥ $x^2 \sin x$

from product rule

$y = uv' + vu'$

$x^2(\cos x) + \sin x(2x)$

$x^2 \cos x + 2x \sin x$

⑦ $y = \cos(2x+5)$

let $u = 2x+5$

$\frac{du}{dx} = 2$

let $y = \cos u$

$\frac{dy}{du} = -\sin u$

$\frac{dy}{dx} = -\sin u \times 2$

$-2\sin u$

put $u = 2x+5$

$-2\sin(2x+5)$

⑧ $y = \sin(2x+7)$

let $u = 2x+7$

$\frac{du}{dx} = 2$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

$\frac{dy}{dx} = \cos u \times 2$

$2\cos u$

put $u = 2x+7$

$2\cos(2x+7)$

⑨ $y = \tan(4x+9)$

let $u = 4x+9$

$\frac{du}{dx} = 4$

let $y = \tan u$; $\frac{dy}{du} = \sec^2 u$

$\frac{dy}{dx} = \sec^2 u \times 4$

$4\sec^2 u$

put $u = 4x+9$

$4\sec^2(4x+9)$

(10) $y = \sin^3 5x \Rightarrow (\sin 5x)^3$

$= (\sin 5x)^3$

let $u = \sin 5x$

$\frac{du}{dx} = 5\cos 5x$

let $y = u^3$

$\frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = 3u^2 \times 5\cos 5x$

$15u^2 \cos 5x$

put $u = \sin 5x$

$15\sin^2 5x \cos 5x$

Note that:

$\rightarrow \sin 2x \neq \sin^2 x$ (linear, power)

$\rightarrow \sin^4 x \neq \sin 4x$ (power, linear)

$\rightarrow \cos^2 x \neq \cos 2x$ (power, linear)

$\rightarrow \cos^5 x \neq \cos 5x$ (power, linear)

$\rightarrow \tan^2 x \neq \tan 2x$

$\rightarrow \tan^4 x \neq \tan 4x$

---

power is different from linear so be careful of their solving pls

$\sin^3 x \neq \sin 3x$ (power, linear)

(11) $y = \sin 7x$

let $u = 7x$

$\frac{du}{dx} = 7$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

$\frac{dy}{dx} = \cos u \times 7$

$7\cos u$

put $u = 7x$

$7\cos 7x$

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(12) $y = \sin^7 x = (\sin x)^7$

let $u = \sin x$

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$\frac{du}{dx} = \cos x$

let $y = u^7$

$\frac{du}{dx} = 7u^6$

$\frac{dy}{dx} = 7u^6 \times \cos x$

$7u^6 \cos x$

put $u = \sin x$

$7\sin^6 x \cos x$

(13)

$y = \frac{\sin 3x}{\cos 3x}$

$\frac{\sin 3x}{\cos 3x} = \tan 3x$

$\Rightarrow y = \tan 3x$

let $u = 3x$

$\frac{du}{dx} = 3$

let $y = \tan u$

$\frac{dy}{du} = \sec^2 u$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \sec^2 u \times 3$

$3\sec^2 u$

put $u = 3x$

$3\sec^2 3x$

(14) $y = \cos(x^2)$

let $u = x^2$

$\frac{du}{dx} = 2x$

let $y = \cos u$

$\frac{dy}{du} = -\sin u$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = -\sin u \times 2x$

$-2x \sin u$

put $u = x^2$

$= -2x \sin x^2$

$-2x \sin x^2$

(15) $y = \sin x^5$

let $u = x^5$

$\frac{du}{dx} = 5x^4$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \cos u \times 5x^4$

$5x^4 \cos u$

Put $u = x^5$

$5x^4 \cos x^5$

---

(16) $y = x^2 \cos x$

from product rule

$x^2(-\sin x) + \cos x (2x)$

$-x^2 \sin x + 2x \cos x$

$2x \cos x - x^2 \sin x$

---

(17) $\sin 4x \log(3x-1)$

from product rule

$\sin 4x \left(\frac{3}{3x-1}\right) + \log(3x-1)(4\cos 4x)$

$= \frac{3 \sin 4x}{3x-1} + 4 \cos 4x \log(3x-1)$

---

(18) $y = \sin(3x^4 + 2x + 1)$

let $u = 3x^4 + 2x + 1$

$\frac{du}{dx} = 12x^3 + 2$

let $y = \sin u$

$\frac{dy}{du} = \cos u$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \cos u \times 12x^3 + 2$

$(12x^3 + 2) \cos u$

put $u = 3x^4 + 2x + 1$

$= (12x^3 + 2) \cos(3x^4 + 2x + 1)$

---

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Differentiation by Inverse of trigonometry (75)

Inverse of trigonometry functions

Function	Derivative
$\sin^{-1}(x)$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}(x)$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1}(x)$	$\frac{1}{1+x^2}$

Examples

(1) $y = \sin^{-1} 5x$

let $u = 5x$

$\frac{du}{dx} = 5$

let $y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times 5 = \frac{5}{\sqrt{1-u^2}}$

put $u = 5x$

$\frac{dy}{dx} = \frac{5}{\sqrt{1-(5x)^2}} = \frac{5}{\sqrt{1-25x^2}}$

(2) $y = \sin^{-1} x^2$

let $u = x^2$

$\frac{du}{dx} = 2x$

let $y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times 2x = \frac{2x}{\sqrt{1-u^2}}$

put $u = x^2$

$\frac{dy}{dx} = \frac{2x}{\sqrt{1-(x^2)^2}} = \frac{2x}{\sqrt{1-x^4}}$

(3) $\cos^{-1}3x$

let $u = 3x$ $\frac{du}{dx} = 3$

let $y = \cos^{-1}u$ $\frac{dy}{du} = \frac{-1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{-1}{\sqrt{1-u^2}} \times 3$

$\frac{-3}{\sqrt{1-u^2}}$

recall that $u = 3x$

$= \frac{-3}{\sqrt{1-(3x)^2}} = \frac{-3}{\sqrt{1-9x^2}}$

(4) $y = \cos^{-1}6x$

let $u = 6x$ $\frac{du}{dx} = 6$

let $y = \cos^{-1}u$ $\frac{dy}{du} = \frac{-1}{\sqrt{1-u^2}}$

$\frac{dy}{dx} = \frac{-1}{\sqrt{1-u^2}} \times 6 = \frac{-6}{\sqrt{1-u^2}}$

$u = 6x$

$\frac{-6}{\sqrt{1-(6x)^2}} = \frac{-6}{\sqrt{1-36x^2}}$

(5) $y = \cos^{-1}x^3$ by shortcut

$= \frac{-3x^2}{\sqrt{1-(x^3)^2}}$

$= \frac{-3x^2}{\sqrt{1-x^6}}$

(6) $y = \tan^{-1}4x$

let $u = 4x$ $\frac{du}{dx} = 4$

let $y = \tan^{-1}u$ $\frac{dy}{du} = \frac{1}{1+u^2}$

$\frac{dy}{dx} = \frac{1}{1+u^2} \times 4 = \frac{4}{1+u^2}$

recall that $u = 4x$

$\frac{4}{1+(4x)^2} = \frac{4}{1+16x^2}$

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(77)

(7) $y = \sin^{-1} ax^2$

let $u = ax^2$

$\frac{du}{dx} = 2x9$

let $y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times 2x9$

$= \frac{2x9}{\sqrt{1-u^2}}$

put $u = ax^2$

$\frac{2x9}{\sqrt{1-(ax^2)^2}} = \frac{2x9}{\sqrt{1-a^2x^4}}$

(8) $y = \tan^{-1}(2x+1)$

let $u = 2x+1$

$\frac{du}{dx} = 2$

let $y = \tan^{-1} u$

$\frac{dy}{du} = \frac{1}{1+u^2}$

by chain rule

$\frac{dy}{dx} = \frac{1}{1+u^2} \times 2$

$\frac{2}{1+u^2}$

recall that

$u = 2x+1$

$= \frac{2}{1+(2x+1)^2}$

expand $(2x+1)(2x+1)$

$= \frac{2}{1+4x^2+4x+1}$

$= \frac{2}{4x^2+4x+2} = \frac{2 \times 1}{2(2x^2+2x+1)}$

$= \frac{1}{2x^2+2x+1}$

$4x^2+2x+2x+1$

$4x^2+4x+1$

Hyperbolic function and inverse of hyperbolic

Examples

(1) $y = \sinh x$

$\frac{dy}{dx} = \cosh x$

(2) $y = \cosh x$

$\frac{dy}{dx} = \sinh x$

(3) $y = \sinh 2x$

Let $u = 2x$

$\frac{du}{dx} = 2$

Let $y = \sinh u$

$\frac{dy}{du} = \cosh u$

by chain rule

$\frac{dy}{dx} = \cosh u \times 2 = 2\cosh u$

put $u = 2x$

$2\cosh 2x$

(4) $y = \cosh 4x$

Let $u = 4x$ $\frac{du}{dx} = 4$

Let $y = \cosh u$ $\frac{dy}{du} = \sinh u$

$\frac{dy}{dx} = \sinh u \times 4$

$= 4 \sinh u$

put $u = 4x$

$4\sinh 4x$

(5) $y = \tanh 5x$

by shortcut

$\frac{dy}{dx} = 5\operatorname{sech}^2 5x$

(6) $y = \sinh 9x$

by shortcut

$9\cosh 9x$

(7) $y = \cosh 100x$

$100\cosh 100x$

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(77)

(7) $y = \sin^{-1} ax^2$

Let $u = ax^2$

$\frac{du}{dx} = 2x a$

Let $y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times 2xa$

$= \frac{2xa}{\sqrt{1-u^2}}$

Put $u = ax^2$

$\frac{2xa}{\sqrt{1-(ax^2)^2}} = \frac{2xa}{\sqrt{1-a^2x^4}}$

(8) $y = \tan^{-1}(2x+1)$

Let $u = 2x+1$

$\frac{du}{dx} = 2$

Let $y = \tan^{-1} u$

$\frac{dy}{du} = \frac{1}{1+u^2}$

by chain rule

$\frac{dy}{dx} = \frac{1}{1+u^2} \times 2$

$\frac{2}{1+u^2}$

recall that

$u = 2x+1$

$= \frac{2}{1+(2x+1)^2}$

expand $(2x+1)(2x+1)$

$= \frac{2}{1+4x^2+4x+1}$

$= \frac{2}{4x^2+4x+2} = \frac{2}{2(2x^2+2x+1)} \times \frac{1}{1}$

$= \frac{1}{2x^2+2x+1}$

$4x^2+2x+2x+1$

$4x^2+4x+1$

Hyperbolic functions and inverse of hyperbolic

mples

= Sinhx

$\frac{dy}{dx}$ = Coshx

y = Coshx

$\frac{dy}{dx}$ = Sinhx

y = Sinh2x

let u = 2x

$\frac{du}{dx}$ = 2

let y = Sinhu

$\frac{dy}{du}$ = Coshu

y chain rule

= Coshu x 2 = 2Coshu

put u = 2x

2Cosh2x

(4) y = Cosh4x

let u = 4x $\frac{du}{dx}$ = 4

let y = Coshu $\frac{dy}{du}$ = Sinhu

$\frac{dy}{dx}$ = Sinhu x 4

= 4 Sinhu

put u = 4x

4Sinh4x

(5) y = tanh5x

by shortcut

$\frac{dy}{dx}$ = 5Sec²h5x

(6) y = Sinh9x

by shortcut

9Cosh9x

(7) y = Cosh100x

100Cosh100x

(78)

Hyperbolic functions and inverse of hyperbolic

Examples

(1) $y = \sinh x$

$\frac{dy}{dx} = \cosh x$

(2) $y = \cosh x$

$\frac{dy}{dx} = \sinh x$

(3) $y = \sinh 2x$

Let $u = 2x$

$\frac{du}{dx} = 2$

Let $y = \sinh u$

$\frac{dy}{du} = \cosh u$

by chain rule

$\frac{dy}{dx} = \cosh u \times 2 = 2\cosh u$

put $u = 2x$

$2\cosh 2x$

(4) $y = \cosh 4x$

Let $u = 4x$ $\frac{du}{dx} = 4$

Let $y = \cosh u$ $\frac{dy}{du} = \sinh u$

$\frac{dy}{dx} = \sinh u \times 4$

$= 4 \sinh u$

put $u = 4x$

$4 \sinh 4x$

(5) $y = \tanh 5x$

by shortcut

$\frac{dy}{dx} = 5\operatorname{sech}^2 5x$

(6) $y = \sinh 9x$

by shortcut

$9\cosh 9x$

(7) $y = \cosh 100x$

$100 \cosh 100x$

Solved and Compiled by Sheriff aka Schrodinger 08108145725

Application of Differentiation

Rate of change

$\text{speed} = \frac{\text{distance}}{\text{time}}$

$\text{velocity} = \frac{\text{displacement}}{\text{time}}$

$\text{acceleration} = \frac{\Delta v}{t} = \frac{v-u}{t}$

$v = \frac{ds}{dt}$ —

$a = \frac{dv}{dt} \Rightarrow \frac{d}{dt}(v) = \frac{d}{dt}\left(\frac{ds}{dt}\right)$

$a = \frac{d^2s}{dt^2}$

Examples

① The motion of a particle described by $S = t^3 + 3t^2 + 4t + 5$ Determined the acceleration of the particle at $t = 4s$

Solution

$S = t^3 + 3t^2 + 4t + 5$

$\text{velocity} = v = \frac{ds}{dt}$

$v = 3t^2 + 6t + 4 + 0$

$v = 3t^2 + 6t + 4$

$a = \frac{dv}{dt} = \frac{d}{dt}(3t^2 + 6t + 4)$

$a = 6t + 6 + 0$

$a = 6t + 6$ —— (*)

at $t = 4s$

$a = 6(4) + 6$

$a = 24 + 6 = 30 ms^{-2}$

② The motion of particle start from rest is described by $S = t^2 + 4t - 3$. find the acceleration of the particle at $t = 1$.

Solution

$v = \frac{ds}{dt} = \frac{d}{dt}(t^2 + 4t - 3)$

$v = 2t + 4$

$a = \frac{dv}{dt} = \frac{d}{dt}(2t + 4)$

$a = 2 + 0 = 2$

$a = 2$

Maximum and minimum values

To find the maximum and minimum values of a function of two variable, the following steps must be taken

(1) identify the function

(2) obtain the first derivative i.e $\frac{dy}{dx}$

(3) At a turning point $\frac{dy}{dx}=0$ Hence we obtain the values which are called Critical value

(4) Obtain the second derivative i.e $\frac{d^2y}{dx^2} \Rightarrow \frac{d^2y}{dx^2}$

(5) The value obtain in third term we substitute into the first

(6) The value obtain is greater than the values is minimum if the value is less than zero it is maximum

$\frac{d^2y}{dx^2} > 0$ = minimum

$\frac{d^2y}{dx^2} < 0$ = maximum

$\frac{d^2y}{dx^2} = 0$ point of inflection

Examples

(1) find the maximum and minimum value of

$y = x^3 - 3x^2 - 9x + 27$

Soln

$y = x^3 - 3x^2 - 9x + 27$

$\frac{dy}{dx} = 3x^2 - 6x - 9 + 0$

$= 3x^2 - 6x - 9$

at $\frac{dy}{dx} = 0$

$3x^2 - 6x - 9 = 0$

$3x^2 - 3x - 3x - 9$

Sum (-3) product (-3)

Solved and compiled by Sheriff a.k.a Schrödinger

$3x^2 - 6x - 9 = 0$

divide through by 3

$\frac{3x^2}{3} - \frac{6x}{3} - \frac{9}{3} = \frac{0}{3}$

$x^2 - 2x - 3 = 0$

$x^2 + x - 3x - 3 = 0$

factorizing

$(x^2 + x) - (3x + 3) = 0$

$x(x+1) - 3(x+1) = 0$

$(x+1)(x-3) = 0$

$(x-3)(x+1) = 0$

at $x - 3 = 0$

$x = 3$

at $x + 1 = 0$

$x = -1$

$x = -1, \quad x = 3$

from $\frac{dy}{dx} = 3x^2 - 6x - 9 = 0$

second derivative

$\frac{d^2y}{dx^2} = 6x - 6$

put the value of x

when $x = -1$

$= 6(-1) - 6$ (81)

$-6 - 6 = -12$

Since $\frac{d^2y}{dx^2} = -12$ hence at $x = -1$ is a maximum point //

Similarly when $x = 3$

put $x = 3$

$\frac{d^2y}{dx^2} = 6(3) - 6$

$= 18 - 6 = 12$

Since $\frac{d^2y}{dx^2} = 12$ Hence at $x = 3$ is a minimum point

when $x = -1$ put it into normal question

$(-1)^3 - 3(-1)^2 - 9(-1) + 27$

$-1 - 3(1) + 9 + 27 = 32$ max value

when $x = 3$

$(3)^3 - 3(3)^2 - 9(3) + 27$

$27 - 27 - 27 + 27$

$0 - 0 = 0$

$0 =$ min value

(2) find the stationary pont of $y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + 5$

Soln

$\frac{dy}{dx} = \frac{3x^2}{3} - \frac{2x}{2} - 2 + 0$

$\frac{dy}{dx} = x^2 - x - 2 + 0$

$\frac{dy}{dx} = x^2 - x - 2$

at $\frac{dy}{dx} = 0$

$x^2 - x - 2 = 0$

$x^2 + x - 2x - 2 = 0$

$x(x+1) - 2(x+1) = 0$

$(x-2)(x+1) = 0$

$x - 2 = 0$

$x = 2$

$x + 1 = 0$

$x = -1$

Hence $x = -1$, $x = 2$ are the Stationary point

(3) find the stationary pont of $y = x^3 - 6x^2 + 9x + 6$

$\frac{dy}{dx} = 3x^2 - 12x + 9$

at $\frac{dy}{dx} = 0$

$3x^2 - 12x + 9 = 0$

$3x^2 - 12x + 9 = 0$

divide through by 3

$\frac{3x^2}{3} - \frac{12x}{3} + \frac{9}{3} = \frac{0}{3}$

$x^2 - 4x + 3 = 0$

$x^2 - x - 3x + 3 = 0$

$x(x-1) - 3(x-1) = 0$

$(x-3)(x-1) = 0$

$x - 3 = 0$

$x = 3$

$x - 1 = 0$

$x = 1$

Hence $x = 3$, $x = 1$ are the stationary pt

General Solutions on Differentiation

find the derivative of the following

(1) $y = x^4 - 2x + \frac{1}{x^2}$

$= x^4 - 2x + x^{-2}$

$\frac{dy}{dx} = 4x^3 - 2 + (-2 \times x^{-2-1})$

$4x^3 - 2 - 2x^{-3}$

$4x^3 - 2 - \frac{2}{x^3}$

(2) $y = \sqrt{x} = x^{1/2}$

$\frac{1}{2}x^{\frac{1}{2}-1}$

$= \frac{1}{2}x^{\frac{1}{2}-\frac{1}{1}} = \frac{1}{2}x^{\frac{1-2}{2}}$

$= \frac{1}{2}x^{-1/2} = \frac{1}{2}x^{1/2}$

$= \frac{1}{2\sqrt{x}}$

(3) $y = \frac{1}{x^3} = x^{-3}$

$= -3 \times x^{-3-1} = -3x^{-4}$

$= \frac{-3}{x^4}$

(4) $y = \frac{1}{x^5} = x^{-5}$

$= -5 \times x^{-5-1} = -5x^{-6}$

$= \frac{-5}{x^6}$

(5) $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

$= x^{1/2} + \frac{1}{x^{1/2}}$

$= x^{1/2} + x^{-1/2}$

$= \frac{1}{2}x^{\frac{1}{2}-1} + \left(-\frac{1}{2} \times x^{-\frac{1}{2}-1}\right)$

$= \frac{1}{2}x^{\frac{1-2}{2}} + \left(-\frac{1}{2}x^{\frac{-1-2}{2}}\right)$

$= \frac{1}{2}x^{-1/2} + \left(-\frac{1}{2}x^{-3/2}\right)$

$= \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2}$

$= \frac{1}{2x^{1/2}} - \frac{1}{2x^{3/2}}$ or

$= \frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{x^3}}$

(6) $y = x + \frac{1}{x}$

$= x + x^{-1}$

$= 1 + (-1 \times x^{-1-1})$

$= 1 + (-x^{-2})$

$= 1 - \frac{1}{x^2}$ or

take the L.C.M

$\frac{x^2 - 1}{x^2}$

(7) $y = 5x^2 + 6x - 8$, find y' at $x = -1$

$= 10x + 6 - 0$

$= 10x + 6$

put at $x = -1$

$10(-1) + 6 = -10 + 6$

$= -4$

(8) $y = 6\sqrt{x^2 + 2x + 1}$

$= (x^2 + 2x + 1)^{1/6}$

$\frac{1}{6} \times (2x+2)(x^2+2x+1)^{\frac{1}{6}-1}$

$\frac{1}{6}(2x+2)(x^2+2x+1)^{\frac{1-6}{6}}$

$= \frac{1}{6}(2x+2)(x^2+2x+1)^{-5/6}$

$= \frac{(x+2)}{3}(x^2+2x+1)^{-5/6}$

$= \frac{(x+2}{3(x^2+2x+1)^{5/6}}$

(9) $y = (1 - 2x^3)^6$

$= 6x - 6x^2(1-2x^3)^{6-1}$

$-3\,6x^2(1-2x^3)^5$

(10) $(2x+1)(2x-1)$

expand the bracket

$4x^2 - 2x + 2x - 1$

$4x^2 - 0 - 1$

$4x^2 - 1$

differentiate it

$= 8x$

(11) $y = \frac{1-3x}{x+3}$

let $u = 1-3x$

$\frac{du}{dx} = -3$

let $v = x+3$

$\frac{dv}{dx} = 1$

by ~~chain~~ quotient rule

$\frac{dy}{dx} = \frac{v\,du - u\,dv}{v^2}$

$\frac{(x+3)(-3) - (1-3x)(1)}{(x+3)^2}$

$\frac{-3x - 9 - 1 + 3x}{(x+3)^2}$

$\frac{dy}{dx} = \frac{-10}{(x+3)^2}$

(12) $y = \frac{1}{x(\sqrt{x})}$

$= \frac{1}{x\, x^{1/2}}$ ⇒ by indice

$= \frac{1}{x^{1+\frac{1}{2}}} = \frac{1}{x^{\frac{2+1}{2}}} = \frac{1}{x^{\frac{3}{2}}}$

$= \frac{1}{x^{3/2}} = x^{-3/2}$

$= \frac{-3}{2} x^{-\frac{3}{2}-1}$

$= \frac{-3}{2} x^{\frac{-3-2}{2}}$

$= \frac{-3}{2} x^{-5/2}$

$= \frac{-3}{2x^{5/2}}$ or $\frac{-3}{2\sqrt{x^5}}$

(13) $y = \sin^3(6x^2+2x+1)$

$= \left(\sin(6x^2+2x+1)\right)^3$

let $u = \sin(6x^2+2x+1)$

$\frac{du}{dx} = (12x+2)\cos(6x^2+2x+1)$

let $y = u^3$ $\frac{dy}{du} = 3u^2$

by chain rule

$\frac{dy}{dx} = 3u^2 \times (12x+2)\cos(6x^2+2x+1)$

put $u = \sin(6x^2+2x+1)$

$(36x+6)\sin(6x^2+2x+1)\cos(6x^2+2x+1)$

$= (36x+6)\sin(6x^2+2x+1)\cos(6x^2+2x+1)$

(14) $y = \sin^{-1}(3x^2)$

let $u = 3x^2$

$\frac{du}{dx} = 6x$

let $y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

by chain rule

$\frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times 6x = \frac{6x}{\sqrt{1-u^2}}$

put $u = 3x^2$

$= \frac{6x}{\sqrt{1-(3x^2)^2}} = \frac{6x}{\sqrt{1-9x^4}}$

(15) $y = e^{ax}$

$\frac{dy}{dx} = a e^{ax}$

(16) $y = \tanh 10x$

$\frac{dy}{dx} = 10 \operatorname{sech}^2 h 10x$

(17) $y = 2x^3 + 7x^2 - 6x - 4$

$\frac{dy}{dx} = 6x^2 + 14x - 8$

(18) $y = x^2 + y^2 = 5$ at $(1, 2)$

$= x^2 + y^2 - 5 = 0$

$\left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}\right) = -\left(\frac{2x + 0 - 0}{0 + 2y - 0}\right) = \frac{-2x}{2y}$

$= \frac{-x}{y}$

put $x = 1, y = 2$

$= \frac{-1}{2}$

(19) find the gradient of the curve $x^3 - y^3 - 4x^2 + 3y = 11x + 4$ at point $(1, 2)$

$= x^3 - y^3 - 4x^2 + 3y - 11x + 4 = 0$

$= \left(\frac{3x^2 + 8x - 11}{-3y^2 + 3}\right)$

put $x = 1, y = 2$

$= \left(\frac{3(1)^2 + 8(1) - 11}{-3(2)^2 + 3}\right)$

$= \left(\frac{3 + 8 - 11}{-12 + 3}\right)$

$= -\frac{16}{9}$

Solved and compiled by Sheriff a.k.a Schrödinger
08108145725

INTEGRATION

This is the inverse of differentiation. The quantity to be integrate is called integrand

Types of integration

1. definite — ⊗
2. indefinite — (xx)

Indefinite integral

This is a type of integral that doesn't have upper and lower limit

Examples

If $\bar{I} = \int x^n dx$

$\bar{I} = \frac{x^{n+1}}{n+1} + c$ ——— (1)

where C is constant of integration

(1) is the general formular for integrations

(1) If $y = \int dx$

$= \int x^0 dx$ because we know that anything raise to power 0 is 1

$= \int x^0 dx = \frac{x^{0+1}}{0+1} + c$

$x + c$

(2) $\bar{I} = \int 2 dx$

$= \int 2x^0 dx = \frac{2x^{0+1}}{0+1} + c$

$= 2x + c$

(3) $\bar{I} = \int 3 dx$

$= \int 3x^0 dx$

$= \frac{3x^{0+1}}{0+1} + c$

$= 3x + c$

(4) $I = \int 2x\,dx$

$= \frac{2x^{1+1}}{1+1} + c = \frac{2x^2}{2} + c$

$= x^2 + c$

(5) $I = \int 4x^2\,dx$

$= \frac{4x^{2+1}}{2+1} + c = \frac{4x^3}{3} + c$

(6) $I = \int x\,dx$

$= \frac{x^{1+1}}{1+1} + c = \frac{x^2}{2} + c$

(7) $I = \int x^2\,dx$

$= \frac{x^{2+1}}{2+1} + c = \frac{x^3}{3} + c$

(8) $I = \int x^5\,dx$

$= \frac{x^{5+1}}{5+1} + c = \frac{x^6}{6} + c$

(9) $I = \int 9\,dx = \int 9x^0\,dx$

$= \frac{9x^{0+1}}{0+1} + c = 9x + c$

(10) $I = \int (2x+1)\,dx$

$= \int 2x\,dx + \int 1\,dx$

$\frac{2x^{1+1}}{1+1} + x + c$

$= \frac{2x^2}{2} + x + c$ or $x^2 + c$

(11) $I = \int (2x^3 + 2x + 1)\,dx$

$= \int 2x^3\,dx + \int 2x\,dx + \int 1\,dx$

$= \int 2x^3\,dx + \int 2x\,dx + \int 1\,dx$

$\frac{2x^{3+1}}{3+1} + \frac{2x^{1+1}}{1+1} + x + c$

$\frac{2x^4}{4} + \frac{2x^2}{2} + x + c$

$= \frac{x^4}{2} + x^2 + x + c$

Solved and compiled by Sheriff a.k.a Schrödinger 08108145125

(12) $I = \int (9x^2 + 6x + 5)\,dx$

$= \int 9x^2\,dx + \int 6x\,dx + \int 5\,dx$

$\frac{9x^3}{3} + \frac{6x^2}{2} + 5x + c$

$= 3x^3 + 3x^2 + 5x + c$

(13) If $\frac{dy}{dx} = 3x^2$ find y.

We first cross multiply

$= dy = 3x^2\,dx$

integrate both sides

$\int dy = \int 3x^2\,dx$

recall the act integral of $\int dx = x$

Similarly $\int dy = y$

$\Rightarrow$

$\int dy = \int 3x^2\,dx$

$y = \frac{3x^{2+1}}{2+1} + c$

$y = \frac{3x^3}{3} + c \Rightarrow y = x^3 + c$

(14) If $\frac{dy}{dx} = 8x$ find y?

we first cross multiply

$= dy = 8x\,dx$

integrate both sides

$\int dy = \int 8x\,dx$

$y = \frac{8x^{1+1}}{1+1} \Rightarrow y = \frac{8x^2}{2} + c$

$y = 4x^2 + c$

(15) If $\frac{dy}{dx} = 12x^2$ find y at point $(1,2)$

We first cross multiply

$dy = 12x^2\,dx$

integrate both sides

$\int dy = \int 12x^2\,dx$

$y = \frac{12x^3}{3} + c \Rightarrow y = 4x^3 + c - (1)$

at the point $(1,2)$

$= x = 1,\ y = 2$

from (1) put $x = 1,\ y = 2$

$2 = 4(1)^3 + c \Rightarrow 2 = 4 + c$

$= 2 - 4 = c \Rightarrow c = -2$

Put $c = -2$ into (1)

$y = 4x^3 + c \Rightarrow y = 4x^3 - 2$

(16) $I = \int \sqrt{x}\,dx = \int x^{1/2}\,dx$

$= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c \Rightarrow \frac{x^{\frac{1+2}{2}}}{\frac{1+2}{2}} + c$

$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c \Rightarrow x^{\frac{3}{2}} \div \frac{3}{2} + c$

$= x^{\frac{3}{2}} \times \frac{2}{3} + c$

$= \frac{2x^{\frac{3}{2}}}{3} + c$

(17) $I = \int 3\sqrt{x}\,dx$

$= \int x^{1/3}\,dx$

$= \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} \Rightarrow \frac{x^{\frac{1+3}{3}}}{\frac{1+3}{3}} + c$

$\Rightarrow \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + c$

$x^{\frac{4}{3}} \times \frac{3}{4} + c \Rightarrow \frac{3x^{\frac{4}{3}}}{4} + c$

(18) $I = \int \frac{1}{x}\,dx$

$= \ln x + c$ (constant integral)

(19) $I = \int \frac{1}{x^2}\,dx \Rightarrow \int x^{-2}\,dx$

$\Rightarrow \frac{x^{-2+1}}{-2+1} + c \Rightarrow \frac{x^{-1}}{-1} + c$

divide through by − sign

$= -x^{-1} + c$

but $x^{-1} = \frac{1}{x}$

$= \frac{-1}{x} + c$

(20) $I = \int \frac{1}{x^3}\,dx$

$= \int x^{-3}\,dx = \frac{x^{-3+1}}{-3+1} + c$

$= \frac{x^{-2}}{-2} + c$

divide through by − sign

$\frac{-x^{-2}}{2} + c$ but $x^{-2} = \frac{1}{x^2}$

$= \frac{-1}{2x^2} + c$ or $\frac{-1}{2x^2} + c$

Solved and compiled by Sheriff a.k.a Schrödinger

(21) $I = \int \frac{1}{x^5} dx$

$= \int x^{-5} dx = \frac{x^{-5+1}}{-5+1} + c$

$= \frac{x^{-4}}{-4} + c$

divide through by − sign

$= -\frac{x^{-4}}{4} + c$

but $x^{-4} = \frac{1}{x^4}$

$= \frac{-1}{4x^4} + c$

(22) $I = \int \frac{8}{x} dx$

$= 8 \int \frac{1}{x} dx$

but $\int \frac{1}{x} dx = \ln x + c$

$8 \times (\ln x + c)$

$= 8 \ln x + c$

(23) $I = \int \frac{6}{x} dx$

$= 6 \int \frac{1}{x} dx$

$6 \ln x + c$

(24) $I = \int \sqrt{x} dx$

$= \int x^{1/2} dx$

$= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}$

$= \frac{x^{3/2}}{3/2} + c$

$= x^{3/2} \times \frac{2}{3} + c$

$= \frac{2x^{3/2}}{3} + c$

(25) If $\frac{dy}{dx} = 3x^2$, find y

$\frac{dy}{dx} = 3x^2$

Cross multiply

$dy = 3x^2 dx$

integrate both sides

$\int dy = \int 3x^2 dx$

$y = \frac{3x^3}{3} + c$

$y = x^3 + c$

(26) If $\frac{dy}{dx} = 8x$, find y?

cross multiply

$dy = 8x\,dx$

integrate both side

$\int dy = \int 8x\,dx$

$y = \frac{8x^2}{2} + c$

$y = 4x^2 + c$

(27) if $\frac{dy}{dx} = 12x^2$ find y at point $(1, 2)$

$\Rightarrow \frac{dy}{dx} = 12x^2$

Cross multiply

$dy = 12x^2\,dx$

integrate both sides

$\int dy = \int 12x^2 dx$

$y = \frac{12x^3}{3} + c$

$y = 4x^3 + c$ ——— (x)

from the boundary condition $(1, 2)$

$\Rightarrow$ $x = 1,\ y = 2$

Substitute $x = 1,\ y = 2$ into (x)

$y = 4x^3 + c$

$\Rightarrow 2 = 4(1)^3 + c$

$2 = 4 + c$

Collect like term

$2 - 4 = c$

$\Rightarrow -2 = c$

$c = -2$

Substitute $c = -2$ into (x)

$y = 4x^3 - 2$

$y = 4x^3 - 2$

Solved and Compiled by Sheriff a.k.a Schrodinger (43)

Integration by Substitution

① $I = \int (2x-1)^2 dx$

Let $u = 2x-1$

$\frac{du}{dx} = 2$,

make dx the subject of the formular

$= \frac{du}{2} = \frac{2}{2} dx$

$dx = \frac{du}{2}$

Substitute $dx = \frac{du}{2}$ into the question

$= \int (u)^2 \cdot \frac{du}{2}$

$= \frac{1}{2} \int (u)^2 du$

integrate both sides

$\frac{1}{2} \left[\frac{u}{3}\right]^3$

$= \frac{u^3}{6} + c$

put $u = 2x-1$

$= \frac{(2x-1)^3}{6} + c$

② $I = \int (2x-7)^3 dx$

Let $u = 2x-7$

$\frac{du}{dx} = 2$

$\frac{du}{1} = \frac{2}{2} dx$

$dx = \frac{du}{2}$

$= \int (u)^3 \frac{du}{2}$

$= \frac{1}{2} \int (u)^3 du$

$\frac{1}{2} \left[\frac{u}{4}\right]^4$

$\frac{u^4}{8} + c$

put $u = 2x-7$

$\frac{(2x-7)^4}{8} + c$

(3) $I = \int (1-4x^2)^2 dx$

Let $u = 1-4x^2$

$\frac{du}{dx} = -8x$

make dx subject of the formular

$du = -8x\,dx$

$\frac{du}{-8x} = \frac{-8x\,dx}{-8x}$

$dx = \frac{-du}{8x}$

Substitute dx into the question

$\int (u)^2 \cdot \left(\frac{-du}{8x}\right)$

$= \frac{-1}{8x}\int (u)^2 du$

$= \frac{-1}{8x}\left[\frac{u}{3}\right]^3$

$-\frac{u^3}{24x} + c$

put $u = 1-4x^2$

$= -\frac{(1-4x^2)^3}{24x} + c$

(94)

(4) $I = \int (4x-3)^2 dx$

Let $u = 4x-3$

$\frac{du}{dx} = 4$

$du = \frac{4\,dx}{4}$

$dx = \frac{du}{4}$

$\int (u)^2 \frac{du}{4}$

$= \frac{1}{4}\int (u)^2 du$

$\frac{1}{4}\left[\frac{u}{3}\right]^3$

$\frac{u^3}{12} + c$

put $u = 4x-3$

$\frac{(4x-3)^3}{12} + c$

(95)

(6) $I = \int (3-2x)^{-5} dx$

Let $u = 3-2x$

$\frac{du}{dx} = -2$

$du = -2\,dx$

$\frac{du}{-2} = \frac{-2\,dx}{-2}$

$dx = -\frac{du}{2}$

$-\int (u)^{-5} \frac{du}{2}$

$= \frac{-1}{2}\int (u)^{-5}\,du$

$\frac{-1}{2}\left[\frac{u^{-5+1}}{-5+1}\right] + c$

$\frac{-1}{2}\left(\frac{u^{-4}}{-4}\right) + c$

$= \frac{u^{-4}}{8} + c$

$= \frac{u^{-4}}{8} + c$

Put $u = 3-2x$

$= \frac{(3-2x)^{-4}}{8} + c$

(6) $I = \int \sqrt{3-4x}\,dx$

$= \int (3-4x)^{1/2}\,dx$

Let $u = 3-4x$

$\frac{du}{dx} = -4$

$\frac{du}{-4} = \frac{-4\,dx}{-4}$

$dx = -\frac{du}{4}$

$-\int (u)^{1/2} \frac{du}{4}$

$= \frac{-1}{4}\int (u)^{1/2}\,du$

$-\frac{1}{4}\left(\frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1}\right) + c = \frac{-1}{4}\left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right)$

$-\frac{1}{4}\left(\frac{u^{3/2}}{3/2}\right) + c$

$= -\frac{u^{3/2}}{2\times 3} + c$

$= \frac{-u^{\frac{3}{2}}}{6} + c$

$= -\frac{(3-4x)^{3/2}}{6} + c$

Constant integral

functions	integral
(1) $\int x^n dx$	$= \frac{x^{n+1}}{n+1} + c$
(2) $\int e^x dx$	$= e^x + c$
(3) $\int \frac{1}{x} dx$	$= \ln x + c \Rightarrow \ln x + c$
(4) $\int \sec^2 x \, dx$	$= \tan x + c \Rightarrow \tan x + c$
(5) $\int \sin x \, dx$	$= -\cos x + c$
(6) $\int \cos x \, dx$	$= \sin x + c$
(7) $\int \sin hx \, dx$	$= \cosh x + c$
(8) $\int \cosh x \, dx$	$= \sin hx + c$
(9) $\int a^x dx$	$= \frac{a^x}{\ln a} + c$
10 $\int e^{kx} dx$	$= \frac{e^{kx}}{k} + c$

Solved and Compiled by Sheriff a.k.a Schrodinger

Integral in form of $\int \frac{f(x)}{f'(x)}\,dx$ (97)

Examples

(1) $I = \int \frac{2x}{x^2-1}\,dx$

let $u = x^2 - 1$

$\frac{du}{dx} = 2x$

make dx the subject of the formulae

$\frac{du}{2x} = \frac{2x\,dx}{2x}$

$dx = \frac{du}{2x}$

Substitute $dx = \frac{du}{2x}$ into the question

$\int \frac{2x}{u} \cdot \frac{du}{2x}$

$= \int \frac{1}{u}\,du$

recall $\int \frac{1}{x}\,dx = \ln x + c$

$= \int \frac{1}{u}\,du = \ln u + c$

put $u = x^2 - 1$

$\ln(x^2-1) + c$

(2) $\int \frac{2x+3}{x^2+3x-5}\,dx$

let $u = x^2 + 3x - 5$

$\frac{du}{dx} = 2x + 3$

make dx the subject of the formulae

$\frac{du}{2x+3} = \frac{(2x+3)\,dx}{2x+3}$

$dx = \frac{du}{2x+3}$

$= \int \frac{2x+3}{u} \cdot \frac{du}{2x+3}$

$= \int \frac{1}{u}\,du \Rightarrow$

$\ln u + c$

recall the

$u = x^2 + 3x - 5$

$\ln(x^2 + 3x - 5) + c$

(3) $\int \frac{3x^2}{x^3-4}\,dx$

let $u = x^3 - 4$

$\frac{du}{dx} = 3x^2$

$\frac{du}{3x^2} = \frac{3x^2\,dx}{3x^2}$

$dx = \frac{du}{3x^2}$

$\int \frac{3x^2}{u} \cdot \frac{du}{3x^2}$

$= \int \frac{1}{u}\,du$

$= \ln u + c$

put $u = x^3 - 4$

$\ln(x^3-4) + c$

(4) $\int \frac{6x^2}{x^3-4}\,dx$

$= 2\int \frac{3x^2}{x^3-4}\,dx$

let $u = x^3 - 4$

$\frac{du}{dx} = 3x^2$

$dx = \frac{du}{3x^2}$

(9e)

$= 2\int \frac{3x^2}{u} \cdot \frac{du}{3x^2}$

$= 2\int \frac{1}{u}\,du$

$2 \ln u + c$

put $u = x^3 - 4$

$2\ln(x^3-4) + c$

(5) $\int \frac{6x^5}{x^6-10}\,dx$

let $u = x^6 - 10$

$\frac{du}{dx} = 6x^5$

$du = 6x^5\,dx$

$\frac{du}{6x^5} = \frac{6x^5\,dx}{6x^5}$

$dx = \frac{du}{6x^5}$

$\int \frac{6x^5}{u} \cdot \frac{du}{6x^5} = \int \frac{1}{u}\,du$

$\ln u + c$

put $u = x^6 - 10$

$\ln u + c$

$= \ln(x^6 - 10) + c$

Solved and Compiled by Sheriff aka Schrodinger

Integral in form of $f(x)f'(x)dx$

(1) $I = \int x^2(x^3+3)\,dx$

let $u = x^3+3$

$\frac{du}{dx} = 3x^2$

make dx the subject

$du = 3x^2\,dx$

$\frac{du}{3x^2} = \frac{3x^2\,dx}{3x^2}$

$dx = \frac{du}{3x^2}$

$\int x^2(u)\cdot\frac{du}{3x^2}$

$= \int \frac{u}{3}\,du = \frac{1}{3}\int (u)\,du$

$= \frac{1}{3}\frac{(u)^2}{2} + c$

$= \frac{u^2}{6} + c$

put $u = x^3+3$

$\frac{(x^3+3)^2}{6} + c$

(2) $I = \int 2x(x^2-1)^4\,dx$

let $u = x^2-1$

$\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$

$\int 2x(u)^4\frac{du}{2x}$

$\int (u)^4\,du$

$= \frac{u^{4+1}}{4+1} + c$

$\frac{u^5}{5} + c$

put $u = x^2-1$

$\frac{(x^2-1)^5}{5} + c$

(3) $I = \int (6x+2)(3x^2+2x+1)^3\,dx$

let $u = 3x^2+2x+1$

$\frac{du}{dx} = 6x+2$

$dx = \frac{du}{6x+2}$

$\int (6x+2)(u)^3\frac{du}{6x+2}$

$\int (u)^3\,du$ integrate

$\frac{u^4}{4} + c$

$\frac{(3x^2+2x+1)^4}{4} + c$

(84)

solved and Consider $\int \frac{1}{\quad} dx$ | $dx = \frac{du}{2}$

Integration by Exponential form (100)

Examples

(1) $I = \int e^x dx$

$= e^x + c$

(2) $I = \int e^{2x} dx$

let $u = 2x$

$\frac{du}{dx} = 2$

$du = 2\,dx$

$\frac{du}{2} = \frac{2\,dx}{2}$

$dx = \frac{du}{2}$

$\int e^u \frac{du}{2} = \frac{1}{2}\int e^u du$

$\frac{1}{2}(e^u) + c$

$\frac{e^u}{2} + c$

put $u = 2x$

$= \frac{e^{2x}}{2} + c$

(3) $\int e^{3x^2} dx$

let $u = 3x^2$

$\frac{du}{dx} = 6x$

$dx = \frac{du}{6x}$

$\int e^u \frac{du}{6x} = \frac{1}{6x}\int e^u du$

$\frac{1}{6x}(e^u) + c$

$\frac{e^u}{6x} + c$

$\frac{e^{3x^2}}{6x} + c$

(4) $I = \int e^{4x^2+2x+1} dx$

let $u = 4x^2+2x+1$

$\frac{du}{dx} = 8x+2$

$du = (8x+2)\,dx$

$\frac{du}{8x+2} = \frac{(8x+2)\,dx}{8x+2}$

$dx = \frac{du}{8x+2}$

$\int e^u \frac{du}{8x+2}$

$= \frac{1}{8x+2}\int e^u du$

$\frac{e^u}{8x+2} + c$

$\frac{e^{4x^2+2x+1}}{5x+2} + c$

(5) $I = \int e^{nx+1} dx$

Let $u = nx+1$

$\frac{du}{dx} = n$

$\frac{du}{n} = \frac{n\,dx}{n}$

$dx = \frac{du}{n}$

$\int e^u \cdot \frac{du}{n} = \frac{1}{n}\int e^u du$

$\frac{1}{n} e^u + c$

put $u = nx+1$

$\frac{e^{nx+1}}{n} + c$

by shortcut

We let the differential of the power be at the denumerator. Then rewrite the whole question and plus C to it.

(6) $I = \int e^{4x} dx$

$= \frac{e^{4x}}{4} + c$

(7) $I = \int e^{\sin x} dx$

$= \frac{e^{\sin x}}{\cos x} + c$

(8) $I = \int e^{5x^2+2x+10} dx$

$\frac{e^{5x^2+2x+10}}{10x+2} + c$

(9) $I = \int e^{5x} dx$

$= \frac{e^{5x}}{5} + c$

(10) $I = \int e^{-x} dx$

$= -e^{-x} + c$

Integration by trigonometry functions

Examples

(1) $I = \int \sin x\,dx$

$= -\cos x + c$

(2) $I = \int \cos x\,dx$

$= \sin x + c$

(3) $I = \int \sin 2x\,dx$

let $u = 2x$

$\frac{du}{dx} = 2$

$\frac{du}{2} = \frac{2\,dx}{2}$

$dx = \frac{du}{2}$

$\int \sin u \cdot \frac{du}{2} = \frac{1}{2}\int \sin u\,du$

$= \frac{1}{2}(-\cos u + c)$

$= -\frac{\cos u}{2} + c$

put $u = 2x$

$= -\frac{\cos 2x}{2} + c$

(4) $\int \cos 5x\,dx$

let $u = 5x$

$\frac{du}{dx} = 5$

$\frac{du}{5} = \frac{5\,dx}{5}$

$dx = \frac{du}{5}$

$\int \cos u \cdot \frac{du}{5} = \frac{1}{5}\int \cos u\,du$

$= \frac{1}{5}(\sin u + c)$

$\frac{\sin u + c}{5}$

put $u = 5x$

$\frac{\sin 5x + c}{5}$

(5) $\int \sec^2 x\,dx$

$= \tan x + c$

(6) $\int \cos(7x+2)\,dx$

let $u = 7x+2$

$\frac{du}{dx} = 7$

$\frac{du}{7} = \frac{7\,dx}{7}$

$dx = \frac{du}{7}$

$\int \cos u \frac{du}{7} = \frac{1}{7}\int \cos u\,du$

$= \frac{1}{7}(\sin u + c)$

$= \frac{\sin u + c}{7}$

put $u = 7x+2$

$= \frac{\sin(7x+2)+c}{7}$

(8) $\int \sin(2x-5)\,dx$

let $u = 2x-5$

$\frac{du}{dx} = 2$

$du = \frac{2\,dx}{2}$

$dx = \frac{du}{2}$

$\int \sin u \frac{du}{2} = \frac{1}{2}\int \sin u\,du$

$= \frac{1}{2}(-\cos u + c)$

$= \frac{-\cos u + c}{2}$

put $u = 2x-5$

$= \frac{-\cos(2x-5)+c}{2}$

(9) $\int x^4 \cos x^5\,dx$

let $u = x^5$

$\frac{du}{dx} = 5x^4$

$du = \frac{5x^4\,dx}{5x^4}$

$dx = \frac{du}{5x^4}$

$\int x^4 \cos u \frac{du}{5x^4}$

$= \frac{1}{5}\int \cos u\,du$

$= \frac{1}{5}\sin u$

$= \frac{\sin}{5}$

Solved and Compiled

(10) $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

let $u = \cos x$

$\frac{du}{dx} = -\sin x$

$du = -\sin x\,dx$

$\frac{du}{-\sin x} = \frac{-\sin x\,dx}{-\sin x}$

$dx = \frac{-du}{\sin x}$

$\int \frac{\sin x}{u} \cdot \frac{-du}{\sin x}$

$= -\int \frac{du}{u} = -\int \frac{1}{u}\,du$

but $\int \frac{1}{u} = \ln u + c$

$= -\ln u + c$

put $u = \cos x$

$= -\ln \cos x + c$

(10.4)

$dx = \frac{du}{7x^6}$

$\int x^6 \sin u \frac{du}{7x^6} = \int \frac{\sin u\,du}{7}$

$= \frac{1}{7}\int \sin u\,du$

$= \frac{1}{7}(-\cos u + c)$

$= \frac{-\cos u}{7} + c$

put $u = x^7$

$= \frac{-\cos x^7}{7} + c$

(12) $\int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx$

let $u = \sin x$ $\frac{du}{dx} = \cos x$

$\frac{du}{\cos x} = \frac{\cos x\,dx}{\cos x}$

$dx = \frac{du}{\cos x}$

$\int \frac{\cos x}{u} \cdot \frac{du}{\cos x} = \int \frac{1}{u}\,du$

$= \ln u + c$ but $u = \sin x$

$= \ln \sin x + c$

$y = \frac{1}{(6-x^2}$

$-\ln u + c$

put $u = \cos x$

$\ln \cos x + c$

d the in
alue the

$y)^2 = 5$

$)^2 = 5$

$\sin x^7 dx$

$u = x^7$

$7x^6$

$x^6 dx$

$\frac{}{7x^6}$